AVAZBEKOV SALOHIDDIN

SAT 1200+

O'ZBEKISTON DAVLAT UNIVERSITETLARI UCHUN TAYYORLOV

SAT *math* 0-1

FOUNDATION BOOK

QISQA MUDDAT ICHIDA SAT
UCHUN KERAKLI MATEMATIKA
VA IZGLIZ TILI BILIMLARI

AVAZBEKOV SALOHIDDIN AZAMJON O’G’LI

SAT 1200+

1-KITOB · ASOSLAR

O’zbekiston davlat universitetlari uchun tayyorlov

Qisqa muddat ichida SAT uchun kerakli matematika
va ingliz tili bilimlari

Mualliflik huquqi

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Kitob haqida

SAT 1200+ — Digital SAT imtihonining matematika bo'limiga tayyorlanish uchun yozilgan uch kitoblik seriya. Har bir kitob talabani asta-sekin yuqori ball tomon olib boradi: **1-kitob** asosiy matematik fundamentni mustahkamlaydi, **2-kitob** SATning asosiy mavzularini (algebra, funksiyalar, statistika, geometriya) to'liq ingliz tilida yoritadi, **3-kitob** esa yuqori ballga erishish uchun murakkab masalalar va tezlik strategiyalarini o'rgatadi.

Ushbu kitob zamonaviy, ma'lumotlarga asoslangan metodika asosida tuzilgan: har bir mavzu College Board'ning rasmiy savollar bazasi tahlili, O'zbekiston maktab dasturi bilan moslik va xalqaro ta'lim tizimlaridagi eng yaxshi amaliyotlarni hisobga olgan holda tanlangan.

Har bir darsning tuzilishi:

1. **Qoida/Formula sahifasi** — mavzuning eng muhim qoidalari bitta sahifada, aniq misollar bilan.
2. **Tushuncha (Concept)** — har bir mavzu ta'rif, yechilgan misol va mashq bilan tushuntiriladi.
3. **Lug'at (Vocabulary)** — SAT savollarida uchraydigan ingliz tilidagi atamalar o'zbekcha tarjimasi bilan.
4. **Mashqlar (Problems)** — bosqichma-bosqich qiyinlashib boruvchi mashqlar to'plami.

Kitob oxirida esa barcha 9 ta darsda uchragan so'zlarni o'z ichiga olgan **to'liq lug'at (complete glossary)** berilgan — bu 2-kitobga o'tishda talaba allaqachon tanish bo'lgan ingliz tilidagi SAT lug'ati bilan qurollangan bo'ladi.

1 Asosiy arifmetika va amallar tartibi

FORMULA SHEET

QOIDA · Amal nomlari

Belgi	Amal	Natija nomi	English
$+$	Qo'shish	Yig'indi	Sum
$-$	Ayirish	Ayirma	Difference
$\times$ / ab	Ko'paytirish	Ko'paytma	Product
$\frac{a}{b}$	Bo'lish	Bo'linma	Quotient

QOIDA · Bajarish tartibi

Har bir amal o'z bosqichiga ega. Bitta misolda barcha to'rt bosqich ketma-ket qanday ishlashini ko'ring:

1. Qavs | **2. Daraja / Ildiz** | **3. × ÷** | **4. + −**

$$5 + 2 \times \underline{(8-3)}^{2} \div 5 \quad \leftarrow \text{1. qavs}$$
$$\downarrow$$
$$5 + 2 \times \underline{5^{2}} \div 5 \quad \leftarrow \text{2. daraja}$$
$$\downarrow$$
$$5 + \underline{2 \times 25 \div 5} \quad \leftarrow \text{3. ×÷ (chapdan o'ngga)}$$
$$\downarrow$$
$$\underline{5 + 10} \quad \leftarrow \text{4. + −}$$
$$\downarrow$$
$$= 15$$

1-rasm. Bitta ifodaning to'rtala bosqich bo'yicha ketma-ket soddalashuvi.

QOIDA · Qavs ochish qoidalari

Qoida	Misol
$-(+a) = -a$	$-(+5) = -5$
$-(-a) = +a$	$-(-53) = +53$
$+(-a) = -a$	$+(-22) = -22$
$+(+a) = +a$	$+(+52) = +52$

1 Asosiy arifmetika va amallar tartibi

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
+	sum	*yig'indi*
−	difference	*ayirma*
×	product	*ko'paytma*
÷	quotient	*bo'linma*
x^2	exponent	*daraja*
x^2	exponents	*daraja*
=	equal	*teng*
$ax + b = c$	equation	*tenglama*
$ax + b = c$	equations	*tenglamalar*
$3x + 2$	expression	*ifoda*
$3x + 2$	expressions	*ifodalar*
–	value	*qiymat*
–	values	*qiymatlar*
+	positive	*musbat*
−	negative	*manfiy*
>	greater	*katta*
–	greatest	*eng katta*
–	error	*xato*
–	errors	*xatolar*
–	following	*quyidagi*
∴	therefore	*demak*
1 → 2 → 3 → 4	order	*tartib*
( )	parentheses	*qavs*
ℤ	integer	*butun son*
ℤ	integers	*butun sonlar*
0	zero	*nol*
−	minus	*minus*
+	plus	*plyus*
–	result	*natija*
Σ	total	*jami*

1-to'plam. *To'rt bosqichli odatni mustahkamlang: qavs → daraja → II bosqich → I bosqich.*

1. $3 + 4 \times 2 =$

2. $(3 + 4) \times 2 =$

3. $20 - 6 \div 3 =$

4. $(18 - 6) \div 3 =$

5. $2 \times 3^2 =$

6. $(2 \times 3)^2 =$

7. $18 \div 3 \times 2 =$

8. $10 - 2 \times 3 + 4 =$

2-to'plam. *Ishorali sonlar berilgan qiymat sifatida – tartibni qo'llashga e'tibor bering.*

1. $-3 + 4 \times 2 =$

2. $(-3 + 4) \times 2 =$

3. $5 - (-2) \times 3 =$

4. $(-2)^2 \times 3 =$

5. $-2^2 \times 3 =$

6. $10 + (-4) \div 2 =$

7. $(10 + (-4)) \div 2 =$

8. $-5 \times (-3) + 2 =$

3-to'plam. *Kasr chizig'ini guruhlash belgisi sifatida qo'llang.*

1. $\dfrac{3+5}{2\times 2} =$

2. $\dfrac{12-4}{2\times 2} =$

3. $\dfrac{2+3\times 4}{7} =$

4. $\dfrac{5^2-1}{3\times 2} =$

5. $\dfrac{6+2}{2+2} =$

6. $\dfrac{20}{2+3} - 1 =$

7. $\dfrac{18-6}{3\times(1+1)} =$

8. $\dfrac{7+9}{4^2-12} =$

4-to'plam. *Modul belgisini guruhlash belgisi sifatida qo'llang.*

1. $|5-9| =$

2. $|5-9|+3 =$

3. $-|5-9| =$

4. $|3-7|\times 2 =$

5. $|-3\times 4| =$

6. $|6-10|-|2-5| =$

7. $10-|4-12| =$

8. $|(-2)^2-9| =$

5-to’plam. *Aralash – haqiqiy test zichligi: daraja, kasr chizig’i va qavs birgalikda.*

1. $\dfrac{2^3+4}{6-4}-1=$

2. $5+\dfrac{(6-2)^2}{4}=$

3. $\dfrac{10-2\times 3}{2^2}=$

4. $3\times(4+2^2)-\dfrac{10}{2}=$

5. $\dfrac{5^2-3\times 5}{2+3}=$

6. $2+3\times(7-2^2)=$

7. $\dfrac{8+|3-7|}{2\times 2}=$

8. $(-3)^2+2\times(5-8)=$

6-to’plam. *Xatoni toping – buzilgan qadamni ayting, to’g’ri javobni bering.*

1. $20\div 4\times 5$ ifodasida ko’rsatilgan yechimda xato bor:
   “1-qadam: $4\times 5=20$. *2-qadam:* $20\div 20=1$.”
   Xato va to’g’ri javob: .....

2. $15-3+4$ ifodasida ko’rsatilgan yechimda xato bor:
   “1-qadam: $3+4=7$. *2-qadam:* $15-7=8$.”
   Xato va to’g’ri javob: .....

3. $\dfrac{10+2}{2\times 3}$ ifodasida ko’rsatilgan yechimda xato bor:
   “Avval $2\times 3=6$, *so’ng* $2\div 6=\frac{1}{3}$, *so’ng* $10+\frac{1}{3}$.”
   Xato va to’g’ri javob: .....

4. -4^2 ifodasida ko’rsatilgan yechimda xato bor:
   “$-4^2=(-4)^2=16$.”
   Xato va to’g’ri javob: .....

5. $3-|5-9|$ ifodasida ko’rsatilgan yechimda xato bor:
   “$3-5-9=-11$.”
   Xato va to’g’ri javob: .....

7-to’plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $|(-3)^2 - 2 \times (5 - 8)| \div 3 =$

2. $\dfrac{4 + |2 - 6|^2}{3 \times 2 - 2} =$

3. $2 \times \left[(10 - 4)^2 \div 3 + 1\right] - 5 =$

4. $\dfrac{|-12| + 3^2}{7 - 4} =$

5. $1 + 2 \times \left[|4 - 10| - (3 - 1)^2\right] =$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	11	2	14
	3	18	4	4
	5	18	6	36
	7	12	8	8
2-to'plam	1	5	2	2
	3	11	4	12
	5	-12	6	8
	7	3	8	17
3-to'plam	1	2	2	2
	3	2	4	4
	5	2	6	3
	7	2	8	4
4-to'plam	1	4	2	7
	3	-4	4	8
	5	12	6	1
	7	2	8	5
5-to'plam	1	5	2	9
	3	1	4	19
	5	2	6	11
	7	3	8	3
7-to'plam	1	5	2	5
	3	21	4	7
	5	5	–	–
6-to'plam	1. Xato 1-qadamda – bo'lish chapda, lekin ko'paytirish avval bajarilgan. To'g'ri: $20 \div 4 = 5$, so'ng $5 \times 5 = 25$.			
	2. Xato 1-qadamda – ayirish chapda, lekin qo'shish avval bajarilgan. To'g'ri: $15-3 = 12$, so'ng $12+4 = 16$.			
	3. Kasr chizig'i suratni ham guruhlaydi. To'g'ri: $\frac{12}{6} = 2$.			
	4. Qavs yo'q – daraja faqat 4 ga tegishli. To'g'ri: $-4^2 = -16$.			
	5. Modul ichidagi amal avval bajariladi. To'g'ri: $5 - 9 = -4$, $\lvert -4 \rvert = 4$, $3 - 4 = -1$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Surat, maxraj, soddalashtirish

surat → $\frac{24}{36}$ ← maxraj

$\xrightarrow{\div 12}$ $\frac{2}{3}$

eng oddiy ko'rinish

QOIDA · Qo'shish / ayirish

Maxrajlar har xil bo'lsa, avval umumiy maxrajga keltiring – faqat shundan keyin suratlarni qo'shing yoki ayiring:

$\frac{1}{4} + \frac{1}{6}$ ← 1. EKUK(4,6)=12

↓

$\frac{3}{12} + \frac{2}{12}$ ← 2. suratlarni qo'shing

↓

$= \frac{5}{12}$

QOIDA · Ko'paytirish

$$\frac{4}{9} \times \frac{3}{8}$$

qisqartiring: 4&8 → ÷4, 3&9 → ÷3

$$= \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

QOIDA · Bo'lish

$$\frac{3}{4} \div \frac{2}{5}$$

teskarisiga ko'paytiring (K-C-F)

$$= \frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$$

QOIDA · Aralash son ↔ noto'g'ri kasr

$3\frac{2}{5}$ $\xrightarrow{3\times5+2}$ $\frac{17}{5}$ $\qquad$ $\frac{23}{6}$ $\xrightarrow{23\div6}$ $3\frac{5}{6}$

Butun son maxrajga ko'paytiriladi, suratga qo'shiladi – maxraj o'zgarmaydi. Teskarisida: suratni maxrajga bo'ling, bo'linma – butun son, qoldiq – yangi surat.

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
$\frac{a}{b}$	fraction	*kasr*
–	numerator / denominator	*surat / maxraj*
–	simplify / reduce	*soddalashtirish / qisqartirish*
–	common denominator	*umumiy maxraj*
–	LCM (least common multiple)	*EKUK (eng kichik umumiy karrali)*
–	GCD (greatest common divisor)	*EKUB (eng katta umumiy bo'luvchi)*
$\frac{b}{a}$	reciprocal	*teskari son*
$2\frac{1}{2}$	mixed number	*aralash son*
$\frac{17}{5}$	improper fraction	*noto'g'ri kasr*
$\times$	multiply	*ko'paytirish*
$\div$	divide	*bo'lish*
$=$	equal	*teng*
$=$	equivalent	*teng kuchli*
–	whole	*butun*
$+$	sum	*yig'indi*
$-$	difference	*ayirma*
$\times$	product	*ko'paytma*
$\div$	quotient	*bo'linma*
$>$	greater	*katta*
–	value	*qiymat*
–	total	*jami*
$\frac{1}{2}$	half	*yarim*
–	remaining	*qolgan*
–	result	*natija*
–	common	*umumiy*

1-to'plam. *Qisqartiring – javobni eng sodda ko'rinishda yozing.*

1. $\frac{8}{12} =$
2. $\frac{15}{25} =$
3. $\frac{18}{24} =$
4. $\frac{21}{14} =$
5. $\frac{36}{48} =$
6. $\frac{50}{75} =$
7. $\frac{28}{35} =$
8. $\frac{100}{40} =$

2-to'plam. *Qo'shing yoki ayiring – avval umumiy maxraj toping.*

1. $\frac{1}{3} + \frac{1}{4} =$
2. $\frac{2}{5} + \frac{1}{3} =$
3. $\frac{3}{4} - \frac{1}{6} =$
4. $\frac{5}{6} - \frac{2}{9} =$
5. $\frac{1}{2} + \frac{2}{3} - \frac{1}{4} =$
6. $\frac{7}{10} - \frac{1}{4} =$
7. $\frac{3}{8} + \frac{5}{12} =$
8. $\frac{11}{12} - \frac{3}{4} =$

3-to'plam. *Ko'paytiring – hisoblashdan oldin qisqartirishga harakat qiling.*

1. $\frac{2}{3} \times \frac{3}{5} =$
2. $\frac{4}{7} \times \frac{7}{8} =$
3. $\frac{5}{6} \times \frac{2}{3} =$
4. $\frac{3}{4} \times \frac{8}{9} =$
5. $\frac{7}{10} \times \frac{5}{14} =$
6. $\frac{2}{9} \times \frac{3}{4} =$
7. $\frac{5}{8} \times \frac{4}{15} =$
8. $\frac{9}{10} \times \frac{5}{6} =$

4-to'plam. *Bo'ling – ikkinchi kasrni teskarisiga aylantirib ko'paytiring.*

1. $\frac{2}{3} \div \frac{4}{5} =$
2. $\frac{3}{4} \div \frac{1}{2} =$
3. $\frac{5}{6} \div \frac{5}{9} =$
4. $\frac{7}{8} \div \frac{7}{4} =$
5. $\frac{2}{5} \div \frac{3}{10} =$
6. $\frac{9}{10} \div \frac{3}{5} =$
7. $\frac{4}{9} \div \frac{8}{3} =$
8. $\frac{5}{12} \div \frac{5}{6} =$

5-to'plam. *Aralash son ↔ noto'g'ri kasr.*

1 $2\frac{3}{4}$ ni noto'g'ri kasrga aylantiring.

2 $5\frac{1}{6}$ ni noto'g'ri kasrga aylantiring.

3 $\frac{17}{5}$ ni aralash songa aylantiring.

4 $\frac{29}{4}$ ni aralash songa aylantiring.

5 $1\frac{1}{2} + 2\frac{1}{3} =$ (aralash son)

6 $3\frac{1}{4} - 1\frac{3}{4} =$ (aralash son)

7 $2\frac{2}{5} \times 1\frac{1}{4} =$

8 $\frac{53}{6}$ ni aralash songa aylantiring.

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1 $\frac{1}{3} + \frac{1}{4}$ ifodasida ko'rsatilgan yechimda xato bor:

"$\frac{1+1}{3+4} = \frac{2}{7}$."

Xato va to'g'ri javob: .....

2 $\frac{2}{5} + \frac{2}{5}$ ifodasida ko'rsatilgan yechimda xato bor:

"Suratni saqlab, maxrajlarni qo'shamiz: $\frac{2}{10}$*."*

Xato va to'g'ri javob: .....

3 $\frac{3}{4} \div \frac{2}{5}$ ifodasida ko'rsatilgan yechimda xato bor:

"$\frac{4}{3} \times \frac{2}{5} = \frac{8}{15}$."

Xato va to'g'ri javob: .....

4 $\frac{16}{24}$ ni soddalashtirishda ko'rsatilgan yechimda xato bor:

"2 ga qisqartirib, $\frac{8}{12}$ *– tayyor."*

Xato va to'g'ri javob: .....

5 $2\frac{3}{4}$ ni noto'g'ri kasrga aylantirishda ko'rsatilgan yechimda xato bor:

"$\frac{2+3}{4} = \frac{5}{4}$."

Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $\left(\frac{1}{2}+\frac{1}{3}\right)\times\frac{2}{5}=$

2. $3\frac{1}{2}\div\left(\frac{1}{4}+\frac{1}{4}\right)=$

3. $\left(\frac{2}{3}-\frac{1}{6}\right)\times\left(\frac{3}{4}+\frac{1}{2}\right)=$

4. $1\frac{3}{4}\times\frac{2}{7}+\frac{1}{8}=$

5. $\left(\frac{5}{6}\div\frac{5}{9}\right)-\left(\frac{1}{2}\times\frac{2}{3}\right)=$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$\frac{2}{3}$	2	$\frac{3}{5}$
	3	$\frac{3}{4}$	4	$\frac{3}{2}$
	5	$\frac{3}{4}$	6	$\frac{2}{3}$
	7	$\frac{4}{5}$	8	$\frac{5}{2}$
2-to'plam	1	$\frac{7}{12}$	2	$\frac{11}{15}$
	3	$\frac{7}{12}$	4	$\frac{11}{18}$
	5	$\frac{11}{12}$	6	$\frac{9}{20}$
	7	$\frac{19}{24}$	8	$\frac{1}{6}$
3-to'plam	1	$\frac{2}{5}$	2	$\frac{1}{2}$
	3	$\frac{5}{9}$	4	$\frac{2}{3}$
	5	$\frac{1}{4}$	6	$\frac{1}{6}$
	7	$\frac{1}{6}$	8	$\frac{3}{4}$
4-to'plam	1	$\frac{5}{6}$	2	$\frac{3}{2}$
	3	$\frac{3}{2}$	4	$\frac{1}{2}$
	5	$\frac{4}{3}$	6	$\frac{3}{2}$
	7	$\frac{1}{6}$	8	$\frac{1}{2}$
5-to'plam	1	$\frac{11}{4}$	2	$\frac{31}{6}$
	3	$3\frac{2}{5}$	4	$7\frac{1}{4}$
	5	$3\frac{5}{6}$	6	$1\frac{1}{2}$
	7	3	8	$8\frac{5}{6}$
7-to'plam	1	$\frac{1}{3}$	2	7
	3	$\frac{5}{8}$	4	$\frac{5}{8}$
	5	$\frac{7}{6}$	–	–
6-to'plam	1. Surat/maxraj mustaqil qo'shilgan, umumiy maxraj topilmagan. To'g'ri: $\frac{1}{3} = \frac{4}{12}$, $\frac{1}{4} = \frac{3}{12}$, yig'indi $\frac{7}{12}$.			
	2. Maxrajlar teng bo'lsa, faqat suratlar qo'shiladi. To'g'ri: $\frac{2}{5} + \frac{2}{5} = \frac{4}{5}$.			
	3. Teskari kasr ikkinchi (bo'luvchi) kasrga qo'llanishi kerak edi. To'g'ri: $\frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$.			
	4. Bitta qisqartirishdan keyin to'xtagan (8 va 12 hali 4 ga bo'linadi). To'g'ri: $\frac{16}{24} = \frac{2}{3}$.			
	5. Butun son maxrajga ko'paytirilmagan. To'g'ri: $3 \times 4 = 12$, $12 + 3 = 15$, ya'ni $\frac{11}{4}$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Nuqtani tekislash

$$\begin{array}{r} 3.70 \\ +2.45 \\ \hline 6.15 \end{array}$$

Qo’shish/ayirishdan oldin nuqtalarni ustma-ust tekislang; yetishmagan xonani nol bilan to’ldiring.

QOIDA · O’nli kasr ↔ oddiy kasr

$$0.75 = \frac{75}{100} = \frac{3}{4} \quad (\div 25,\ \text{EKUB}) \qquad \frac{1}{8} \xrightarrow{\div} 0.125$$

Oxirgi xona qiymati maxrajni belgilaydi; teskarisida suratni maxrajga bo’ling.

QOIDA · Foiz ↔ o’nli kasr

$$45\% \xrightarrow{\div 100} 0.45 \qquad 0.6 \xrightarrow{\times 100} 60\%$$

Foizdan o’nli kasrga: vergulni 2 xona chapga suring. Teskarisida: 2 xona o’ngga.

QOIDA · Sonning foizi

80 ning 25%i

↓ 1. 25% = 0.25

0.25×80

↓ 2. ko’paytiring

$= 20$

VOCABULARY – LUG’AT

Belgi	English	O’zbekcha
%	**percent**	*foiz*
%	**percentage**	*foiz*
%	**percentages**	*foizlar*
–	**decimal**	*o’nlik*
$\frac{a}{b}$	**fraction**	*kasr*
$\frac{a}{b}$ (top)	**numerator**	*surat*
$\frac{a}{b}$ (bottom)	**denominator**	*maxraj*
=	**equivalent**	*teng kuchli*
=	**equal**	*teng*
–	**value**	*qiymat*
–	**values**	*qiymatlar*
–	**whole**	*butun*
>	**greater**	*katta*
≈	**approximately**	*taxminan*
–	**nearest**	*eng yaqin*
–	**round**	*yaxlitlamoq*
×	**multiplying**	*ko’paytirish*
×	**multiplied**	*ko’paytirildi*
÷	**dividing**	*bo’lish*
÷	**divided**	*bo’lingan*
–	**difference**	*ayirma*
+	**sum**	*yig’indi*
–	**subtracting**	*ayirish*
+	**adding**	*qo’shish*
–	**number**	*son*
–	**numbers**	*sonlar*
–	**total**	*jami*
–	**simplifying**	*soddalashtirish*
.	**decimal point**	*vergul (sat-da: nuqta)*
–	**place value**	*xona qiymati*
–	**to align**	*tekislamoq*
–	**to convert**	*aylantirmoq*
gcd	**greatest common divisor**	*eng katta umumiy bo’luvchi*
–	**percent of a number**	*sonning foizi*

1-to’plam. *O’nli kasrlarni qo’shish va ayirish: vergulni tekislashni unutmang.*

1. $3.7 + 2.45 =$

2. $12.6 - 4.75 =$

3. $0.8 + 0.09 =$

4. $15 - 3.25 =$

5. $2.05 + 1.95 =$

6. $100 - 0.4 =$

7. $0.125 + 0.875 =$

8. $7.3 - 2.08 =$

2-to’plam. *O’nli kasrlarni ko’paytirish va bo’lish – kasr xonalarini to’g’ri sanang.*

1. $0.3 \times 0.4 =$

2. $2.5 \times 1.2 =$

3. $0.06 \times 5 =$

4. $1.44 \div 1.2 =$

5. $0.5 \times 0.5 =$

6. $7.2 \div 0.4 =$

7. $0.02 \times 0.03 =$

8. $9.6 \div 2.4 =$

3-to’plam. *O’nli kasr ↔ oddiy kasr almashtirish.*

1. 0.5 ni oddiy kasrga aylantiring.

2. 0.25 ni oddiy kasrga aylantiring.

3. 0.75 ni oddiy kasrga aylantiring.

4. 0.2 ni oddiy kasrga aylantiring.

5. $\frac{3}{4}$ ni o’nli kasrga aylantiring.

6. $\frac{1}{8}$ ni o’nli kasrga aylantiring.

7. $\frac{7}{10}$ ni o’nli kasrga aylantiring.

8. 0.125 ni oddiy kasrga aylantiring.

4-to’plam. *Foiz ↔ o’nli kasr almashtirish – 100% dan katta va 1% dan kichik holatlarni ham o’z ichiga oladi.*

1. 45% ni o’nli kasrga aylantiring.

2. 7% ni o’nli kasrga aylantiring.

3. 150% ni o’nli kasrga aylantiring.

4. 0.5% ni o’nli kasrga aylantiring.

5. 0.6 ni foizga aylantiring.

6. 1.25 ni foizga aylantiring.

7. 0.03 ni foizga aylantiring.

8. 2 ni foizga aylantiring.

5-to'plam. *Foiz ↔ oddiy kasr, va sonning foizini topish.*

1. 80 ning 25% i qanchaga teng?

2. 150 ning 10% i qanchaga teng?

3. 45 ning 60% i qanchaga teng?

4. $\frac{1}{4}$ ni foizga aylantiring.

5. $\frac{3}{5}$ ni foizga aylantiring.

6. 150% ni oddiy kasrga aylantiring.

7. 50 ning 12% i qanchaga teng?

8. 20 ning 90% i qanchaga teng?

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. 80 ning 45% ini hisoblashda ko'rsatilgan yechimda xato bor:

   "$45 \times 80 = 3600$, *demak javob* 3600."

   Xato va to'g'ri javob: .....

2. $3.7 + 2.45$ ni hisoblashda ko'rsatilgan yechimda xato bor:

   "Vergulni tekislamasdan qo'shib, $3.7 + 2.45 = 2.82$ *deb topilgan."*

   Xato va to'g'ri javob: .....

3. $1.44 \div 1.2$ ni hisoblashda ko'rsatilgan yechimda xato bor:

   "Kasr xonalarini noto'g'ri sanab, $1.44 \div 1.2 = 0.12$ *deb topilgan."*

   Xato va to'g'ri javob: .....

4. 0.2 ni oddiy kasrga aylantirishda ko'rsatilgan yechimda xato bor:

   "0.2 *dagi raqam yuzdan bir xonasida deb o'ylab,* $\frac{2}{100} = \frac{1}{50}$ *deb topilgan."*

   Xato va to'g'ri javob: .....

5. 150% ni o'nli kasrga aylantirishda ko'rsatilgan yechimda xato bor:

   "Vergul faqat bir xona chapga surilib, $150\% = 15$ *deb topilgan."*

   Xato va to'g'ri javob: .....

7-to’plam. *Bonus / qiyin – aralash almashtirishlar, har biriga taxminan 90 soniya.*

1 0.8 ning 35% ini toping, so’ngra natijani 100 ga ko’paytiring.

2 $\frac{3}{8}$ ni foizga aylantiring.

3 0.6 ning necha foizi 0.3 ga teng?

4 1.5 ni foizga va oddiy kasrga aylantiring.

5 (0.25 + 0.5) yig’indisining 40% ini toping.

Javoblar

To’plam	#	Javob	#	Javob
1-to’plam	1	6.15	2	7.85
	3	0.89	4	11.75
	5	4	6	99.6
	7	1	8	5.22
2-to’plam	1	0.12	2	3
	3	0.3	4	1.2
	5	0.25	6	18
	7	0.0006	8	4
3-to’plam	1	1/2	2	1/4
	3	3/4	4	1/5
	5	0.75	6	0.125
	7	0.7	8	1/8
4-to’plam	1	0.45	2	0.07
	3	1.5	4	0.005
	5	60%	6	125%
	7	3%	8	200%
5-to’plam	1	20	2	15
	3	27	4	25%
	5	60%	6	3/2
	7	6	8	18
7-to’plam	1	28	2	37.5%
	3	50%	4	150%, 3/2
	5	0.3	–	–
6-to’plam	1. Foiz to’g’ridan-to’g’ri ko’paytirilgan, o’nli kasrga aylantirilmagan. To’g’ri: $45\% = 0.45$, so’ng $0.45 \times 80 = 36$.			
	2. Vergullar tekislanmagan. To’g’ri: $3.70 + 2.45 = 6.15$.			
	3. Kasr xonalari noto’g’ri sanalgan. To’g’ri: $1.44 \div 1.2 = 1.2$.			
	4. Raqamning xonasi noto’g’ri aniqlangan (yuzdan emas, o’ndan). To’g’ri: $0.2 = \frac{2}{10} = \frac{1}{5}$.			
	5. Vergul bir xona o’rniga ikki xona chapga surilishi kerak edi. To’g’ri: $150\% = 1.50 = 1.5$.			

Video yechimlar va qo’shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Musbat/manfiy va modul

Son	Modul	Qarama-qarshi son
7	$\|7\| = 7$	-7
-7	$\|-7\| = 7$	7
-12	$\|-12\| = 12$	12

QOIDA · Qo'shish / ayirish

Bir xil ishora: modullarni qo'shing, ishorani saqlang. Har xil ishora: kattasidan kichigini ayiring, kattaning ishorasini oling. Manfiyni ayirish – musbatini qo'shish: $a - (-b) = a + b$.

$(-7) + 12 - (-5)$ ← 1. $-(-5) = +5$

↓

$(-7) + 12 + 5$ ← 2. chapdan o'ngga

↓

$= 10$

QOIDA · Ko'paytirish / bo'lish

Ishoralar	Natija
bir xil	+
har xil	–

QOIDA · Necha marta manfiy?

$(-4) \times (-3) \div (-2)$

$(-4) \times (-3) = 12$ (bir xil)

$12 \div (-2) = -6$ (har xil)

$= -6$

QOIDA · Qavs ochish

$-(a + b) = -a - b$ $\quad -(a - b) = -a + b$

$-(7 - 4) = -7 + 4$

$-(7 - 4) + 3$

↓

$-7 + 4 + 3$ $\quad = 0$

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
+	**positive**	*musbat*
–	**negative**	*manfiy*
–	**opposite**	*qarama-qarshi*
–	**opposite numbers**	*qarama-qarshi sonlar*
$\|x\|$	**absolute value**	*modul*
–	**sign**	*ishora*
–	**same sign**	*bir xil ishora*
–	**different sign**	*har xil ishora*
0	**zero**	*nol*
–	**number line**	*sonlar o'qi*
=	**equal**	*teng*
>	**greater**	*katta*
–	**value**	*qiymat*
–	**values**	*qiymatlar*
$\mathbb{Z}$	**integer**	*butun son*
–	**integers**	*butun sonlar*
–	**term**	*had*
–	**terms**	*hadlar*
–	**even**	*juft*
–	**odd**	*toq*
–	**result**	*natija*
–	**results**	*natijalar*
×	**multiply**	*ko'paytirmoq*
–	**multiplying**	*ko'paytirish*
÷	**dividing**	*bo'lish*
–	**number**	*son*
–	**numbers**	*sonlar*
–	**distance**	*masofa*
–	**true**	*to'g'ri*
–	**false**	*noto'g'ri*

1-to'plam. *Bir xil ishorali sonlarni qo'shing.*

1. $(-3) + (-8) =$
2. $(-15) + (-9) =$
3. $7 + 12 =$
4. $(-20) + (-5) =$
5. $(-11) + (-14) =$
6. $25 + 16 =$
7. $(-30) + (-12) =$
8. $9 + 18 =$

2-to'plam. *Aralash ishoralar va manfiyni ayirish.*

1. $8 - (-3) =$
2. $(-5) - (-12) =$
3. $(-7) + 10 =$
4. $14 - 20 =$
5. $(-9) - 6 =$
6. $6 - (-6) =$
7. $(-18) + 11 =$
8. $(-4) - (-4) =$

3-to'plam. *Ko'paytirish va bo'lish (ikki ko'paytuvchi).*

1. $(-6) \times 7 =$

2. $(-9) \times (-8) =$

3. $45 \div (-5) =$

4. $(-56) \div (-7) =$

5. $12 \times (-11) =$

6. $(-100) \div 4 =$

7. $(-13) \times (-3) =$

8. $(-81) \div (-9) =$

4-to'plam. *Ko'p ko'paytuvchili zanjirlar – manfiylar sonini sanang.*

1. $(-2) \times (-3) \times (-4) =$

2. $(-5) \times 6 \times (-2) =$

3. $100 \div (-5) \div (-2) =$

4. $(-1) \times (-1) \times (-1) \times (-1) =$

5. $(-3) \times (-3) \times (-3) =$

6. $48 \div (-4) \times (-3) =$

7. $(-10) \times (-10) \div 5 =$

8. $(-2) \times 3 \times (-4) \times (-1) =$

5-to'plam. *Qavslarni oching.*

1. $-(9-4)+6=$

2. $-(5+3)-2=$

3. $10-(6-11)=$

4. $-(-7+2)+1=$

5. $-(8-8)-5=$

6. $3-(-4-6)=$

7. $-(12-20)-9=$

8. $-(-6-(-2))+4=$

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. $8-(-3)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: $8-3=5$*."*

   Xato va to'g'ri javob: .....

2. $(-5)\times(-6)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: -30*."*

   Xato va to'g'ri javob: .....

3. $-(10-4)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: $-10-4=-14$*."*

   Xato va to'g'ri javob: .....

4. $(-2)\times(-2)\times(-2)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: barcha ko'paytuvchilar manfiy bo'lgani uchun javob musbat: 8*."*

   Xato va to'g'ri javob: .....

5. $(-15)\div(-3)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: -5*."*

   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $-(-8+3)\times(-2)-5=$

2. $(-4)\times(-3)-(-7)+(-2)\times 5=$

3. $18\div(-3)+(-2)\times(-4)-6=$

4. $-(5-(-9))+(-14)\div 7=$

5. $(-6)\times[(-2)-3]+(-10)\div(-5)=$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	-11	2	-24
	3	19	4	-25
	5	-25	6	41
	7	-42	8	27
2-to'plam	1	11	2	7
	3	3	4	-6
	5	-15	6	12
	7	-7	8	0
3-to'plam	1	-42	2	72
	3	-9	4	8
	5	-132	6	-25
	7	39	8	9
4-to'plam	1	-24	2	60
	3	10	4	1
	5	-27	6	36
	7	20	8	-24
5-to'plam	1	1	2	-10
	3	15	4	6
	5	-5	6	13
	7	-1	8	8
7-to'plam	1	-15	2	9
	3	-4	4	-16
	5	32	–	–
6-to'plam	1. Minus-minus (manfiyni ayirish) qo'shishga aylantirilmadi. To'g'ri: $8-(-3)=8+3=11$.			
	2. Bir xil ishorali ko'paytirish musbat natija beradi. To'g'ri: $(-5)\times(-6)=30$.			
	3. Manfiy ishora qavs ichidagi faqat birinchi hadga qo'llanildi. To'g'ri: $-(10-4)=-10+4=-6$.			
	4. Uchta (toq sondagi) manfiy ko'paytuvchi bor – toq son manfiylar natijani manfiy qiladi. To'g'ri: $(-2)\times(-2)\times(-2)=-8$.			
	5. Bo'lishda ham bir xil ishora musbat natija beradi. To'g'ri: $(-15)\div(-3)=5$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

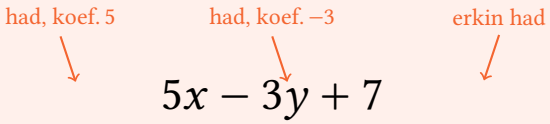
YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Had, koeffitsient, o'xshash hadlar



$4x + 3y - x + 7$ —(x hadlarni qo'shing)→ $3x + 3y + 7$

QOIDA · Taqsimot xossasi

$$a(b + c) = ab + ac$$

$3(2x - 5) = 6x - 15$

Manfiy ishora qavsdagi **har bir** hadga tatbiq etiladi:

$-(3x - 5) \neq -3x - 5$ $\qquad -(3x - 5) = -3x + 5$

QOIDA · Ko'p hadli ifodani soddalashtirish

$4(x + 3) - 2(x - 1)$ ← 1. qavslarni oching

↓

$4x + 12 - 2x + 2$ ← 2. o'xshash hadlarni qo'shing

↓

$= 2x + 14$

Har bir qavs taqsimot xossasi bilan alohida ochiladi (manfiy ishoraga e'tibor bering), keyin o'xshash hadlar ixchamlanadi.

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
x	variable	*o'zgaruvchi*
x, y	variables	*o'zgaruvchilar*
–	term	*had*
–	terms	*hadlar*
$5x$	coefficient	*koeffitsient*
–	coefficients	*koeffitsientlar*
7	constant term	*erkin had*
c	constant	*doimiy*
–	like terms	*o'xshash hadlar*
–	combine like terms	*o'xshash hadlarni ixchamlash*
$a(b+c) = ab+ac$	distributive property	*taqsimot xossasi*
–	expression	*ifoda*
–	expressions	*ifodalar*
–	simplify	*soddalashtirish*
–	simplifies	*soddalashtiradi*
–	simplifying	*soddalashtirish*
=	equivalent	*teng kuchli (ekvivalent)*
( )	parentheses	*qavs*
–	negative	*manfiy*
+	positive	*musbat*
–	minus	*minus*
+	plus	*plyus*
–	value	*qiymat*
–	values	*qiymatlar*
=	equal	*teng*
–	error	*xato*
–	errors	*xatolar*
–	following	*quyidagi*
–	given	*berilgan*
–	represents	*ifodalaydi*

1-to'plam. *O'xshash hadlarni qo'shing (musbat koeffitsientlar).*

1. $3x + 5x =$

2. $7y + 2y + y =$

3. $4a + 3b + 2a =$

4. $6x + 3 + 2x =$

5. $5m + 4n + 3m + n =$

6. $2x + 3x + 4x =$

7. $8p + 5q + 2p + 3q =$

8. $x + x + x + x =$

2-to'plam. *O'xshash hadlarni qo'shing – ayirish ishtirok etadi.*

1. $7x - 3x =$

2. $5a - 8a =$

3. $4x - 7 + 2x =$

4. $-3y + 5y - 2y =$

5. $6m - 4n - 2m + 7n =$

6. $-5x + 3x - 8 =$

7. $2a - 5a + 6a =$

8. $9x - 4x - x =$

3-to'plam. *Taqsimot qonunini qo'llang (musbat ko'paytuvchi).*

1. $3(x+4) =$
2. $5(2x-3) =$
3. $2(3a+4b) =$
4. $4(x-1) =$
5. $6(2+3x) =$
6. $7(x+y) =$
7. $3(4x-2) =$
8. $2(5x+3) =$

4-to'plam. *Taqsimot qonuni – manfiy ko'paytuvchini har bir hadga tarqating.*

1. $-2(x+3) =$
2. $-(3x-5) =$
3. $-4(2x-1) =$
4. $-(x+7) =$
5. $-3(4-x) =$
6. $-(2x-3y) =$
7. $-5(x+2) =$
8. $-(-x+4) =$

5-to'plam. *Aralash – avval taqsimlang, so'ng o'xshash hadlarni qo'shing.*

1. $3(x+2)+4x=$

2. $5(2x-1)-3x=$

3. $2(x+3)-(x-1)=$

4. $4(x-2)+3(x+1)=$

5. $-(2x+3)+5x=$

6. $3(x-1)-2(x+2)=$

7. $2(3x+1)-(4x-5)=$

8. $5-2(x-3)=$

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri soddalashtirilgan ko'rinishni bering.*

1. $-(3x-5)$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$-3x-5$"

   Xato va to'g'ri javob: .....

2. $4x+3x^2$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$7x^2$"

   Xato va to'g'ri javob: .....

3. $-2(x-4)$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$-2x-8$"

   Xato va to'g'ri javob: .....

4. $5x-(2x-3)$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$5x-2x-3=3x-3$"

   Xato va to'g'ri javob: .....

5. $3(2x+1)-4$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$6x+1-4=6x-3$"

   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – ichma-ich qavslar va ikki o'zgaruvchili ifodalar.*

1 $2\,[3(x+1)-2x]+5=$

2 $-3(x-2)+2(x+4)-5=$

3 $4x-2\,[x-(3x+1)]=$

4 $5(a-2b)-3(2a-b)=$

5 $-\,[2(x+3)-(x-1)]+4x=$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$8x$	2	$10y$
	3	$6a + 3b$	4	$8x + 3$
	5	$8m + 5n$	6	$9x$
	7	$10p + 8q$	8	$4x$
2-to'plam	1	$4x$	2	$-3a$
	3	$6x - 7$	4	0
	5	$4m + 3n$	6	$-2x - 8$
	7	$3a$	8	$4x$
3-to'plam	1	$3x + 12$	2	$10x - 15$
	3	$6a + 8b$	4	$4x - 4$
	5	$18x + 12$	6	$7x + 7y$
	7	$12x - 6$	8	$10x + 6$
4-to'plam	1	$-2x - 6$	2	$5 - 3x$
	3	$4 - 8x$	4	$-x - 7$
	5	$3x - 12$	6	$-2x + 3y$
	7	$-5x - 10$	8	$x - 4$
5-to'plam	1	$7x + 6$	2	$7x - 5$
	3	$x + 7$	4	$7x - 5$
	5	$3x - 3$	6	$x - 7$
	7	$2x + 7$	8	$11 - 2x$
7-to'plam	1	$2x + 11$	2	$9 - x$
	3	$8x + 2$	4	$-a - 7b$
	5	$3x - 7$	–	–
6-to'plam	1. Xato – manfiy ishora faqat $3x$ ga tarqatilgan, -5 ga emas. To'g'ri: $-(3x - 5) = 5 - 3x$.			
	2. Xato – x va x^2 o'xshash had emas (daraja boshqa), qo'shib bo'lmaydi. To'g'ri: $3x^2 + 4x$ (allaqachon sodda).			
	3. Xato – $-2 \times (-4) = +8$ bo'lishi kerak edi, ishora tushib qolgan. To'g'ri: $-2(x - 4) = 8 - 2x$.			
	4. Xato – qavs oldidagi manfiy ishora faqat $2x$ ga tarqatilgan, -3 ga emas. To'g'ri: $5x - (2x - 3) = 3x + 3$.			
	5. Xato – 3×1 hisoblanmagan (ikkinchi hadga tarqatish o'tkazib yuborilgan). To'g'ri: $3(2x + 1) - 4 = 6x - 1$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Bir qadamli

$$x + 7 = 12$$
$$\downarrow \quad \leftarrow -7 \text{ ikkala tomonga}$$
$$x = 5$$

QOIDA · Ikki qadamli

$$3x - 5 = 10$$
$$\downarrow \quad \leftarrow +5$$
$$3x = 15$$
$$\downarrow \quad \leftarrow \div 3$$
$$x = 5$$

QOIDA · Ko’p qadamli: qavs ochish

Avval qavsni oching va o’xshash hadlarni ixchamlashtiring – shundan keyin oddiy ikki qadamli tenglama qoladi:

$$2(x - 4) + 5 = 7 \quad \leftarrow \text{1. qavs och, hadlarni qo’sh}$$
$$\downarrow$$
$$2x - 3 = 7 \quad \leftarrow \text{2. } +3 \text{ ikkala tomonga}$$
$$\downarrow$$
$$2x = 10 \quad \leftarrow \text{3. } \div 2 \text{ ikkala tomonga}$$
$$\downarrow$$
$$x = 5$$

QOIDA · Ikkala tomonda noma’lum

Noma’lumli hadlarni bir tomonga, sonli hadlarni ikkinchi tomonga yig’ing:

$$5x - 2 = 2x + 7 \quad \leftarrow \text{1. } -2x \text{ ikkala tomondan}$$
$$\downarrow$$
$$3x - 2 = 7 \quad \leftarrow \text{2. } +2 \text{ ikkala tomonga}$$
$$\downarrow$$
$$3x = 9 \quad \leftarrow \text{3. } \div 3 \text{ ikkala tomonga}$$
$$\downarrow$$
$$x = 3$$

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
=	equation	*tenglama*
–	equations	*tenglamalar*
–	solution	*yechim*
–	solutions	*yechimlar*
–	solve	*yechmoq*
–	solving	*yechish*
=	equal	*teng*
–	value	*qiymat*
–	values	*qiymatlar*
x	variable	*o'zgaruvchi*
–	variables	*o'zgaruvchilar*
–	coefficient	*koeffitsient*
–	coefficients	*koeffitsientlar*
–	constant	*doimiy*
–	constants	*o'zgarmaslar*
+	positive	*musbat*
–	negative	*manfiy*
–	error	*xato*
–	errors	*xatolar*
–	term	*had*
–	terms	*hadlar*
( )	parentheses	*qavs*
–	distributive property	*distributiv xossa*
–	given	*berilgan*
–	following	*quyidagi*
–	therefore	*demak*
–	true	*to'g'ri*
–	false	*noto'g'ri*
$\frac{a}{b}$	fraction	*kasr*
–	check (the answer)	*tekshirish*

1-to'plam. *Bir qadamli tenglamalar – asosiy amalni teskarisi bilan bajaring.*

1. $x + 7 = 12$
2. $x - 9 = 4$
3. $5x = 35$
4. $\frac{x}{4} = 6$
5. $x - (-3) = 10$
6. $-3x = 21$
7. $x + 15 = 8$
8. $\frac{x}{-5} = -6$

2-to'plam. *Ikki qadamli tenglamalar – qo'shish/ayirishdan so'ng ko'paytirish/bo'lish.*

1. $2x + 3 = 11$
2. $3x - 5 = 10$
3. $\frac{x}{2} + 4 = 9$
4. $5 - 2x = 1$
5. $4x + 7 = -5$
6. $-3x - 2 = 10$
7. $\frac{x}{3} - 1 = -4$
8. $7 - \frac{x}{2} = 3$

3-to'plam. *Qavsni ochish va o'xshash hadlarni birlashtirish.*

1. $3(x+2)=15$
2. $2(x-4)+5=7$
3. $4x+2x-3=15$
4. $-2(x+3)=10$
5. $3(2x-1)-2x=13$
6. $6-3(x-2)=0$
7. $2(3x+1)-(x-5)=22$
8. $-4(x-1)+2x=-2$

4-to'plam. *O'zgaruvchi ikkala tomonda – avval bir tomonga yig'ing.*

1. $3x+5=x+13$
2. $5x-2=2x+7$
3. $4x+1=2x-9$
4. $7-x=2x+1$
5. $2(x+3)=x+10$
6. $3x-4=5(x-2)$
7. $x+8=3x-2$
8. $6-2x=4x-18$

5-to'plam. *Aralash, SAT uslubida – ba'zilari x emas, hosila ifodani so'raydi. Diqqat bilan o'qing.*

1. $2x + 5 = 17$. $x + 3$ ning qiymati nechaga teng?

2. $3(x - 2) = 2x + 1$. $2x$ ning qiymati nechaga teng?

3. $\frac{x}{2} + 3 = 9$. $3x$ ning qiymati nechaga teng?

4. $4x - 7 = x + 8$. $x - 1$ ning qiymati nechaga teng?

5. $2(2x + 1) - 3 = 15$. x ning qiymati nechaga teng?

6. $5x + 2 = 3x + 12$. $5x$ ning qiymati nechaga teng?

7. $6 - x = 2x - 6$. $x + 4$ ning qiymati nechaga teng?

8. $3(x - 1) = x + 5$. $2x - 1$ ning qiymati nechaga teng?

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. $x - 5 = 12$ ifodasida ko'rsatilgan yechimda xato bor:

   "1-qadam: ikkala tomonga 5 qo'shiladi: $x = 12$."

   Xato va to'g'ri javob: .....

2. $2x - 5 = 9$ ifodasida ko'rsatilgan yechimda xato bor:

   "1-qadam: $2x = 9 - 5 = 4$."

   Xato va to'g'ri javob: .....

3. $-2(x - 3) = 10$ ifodasida ko'rsatilgan yechimda xato bor:

   "1-qadam: $-2x - 6 = 10$."

   Xato va to'g'ri javob: .....

4. $3x + 5 = 20$ ifodasida ko'rsatilgan yechimda xato bor:

   "1-qadam: $8x = 20$, shuning uchun $x = 2.5$."

   Xato va to'g'ri javob: .....

5. $2x+1 = 11$ (savolda $3x$ so'ralgan) ifodasida ko'rsatilgan yechimda xato bor:

   "$x = 5$, javob: 5."

   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – kasr va qavslar birgalikda, har biriga taxminan 90 soniya.*

1. $\frac{x+3}{2} = 7$

2. $\frac{3(x-1)}{2} = 6$

3. $2(x+1) - 3(x-2) = 10$

4. $\frac{2x-1}{3} + 1 = 4$

5. $4(x-2) = 2(x+3) - 6$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	5	2	13
	3	7	4	24
	5	7	6	-7
	7	-7	8	30
2-to'plam	1	4	2	5
	3	10	4	2
	5	-3	6	-4
	7	-9	8	8
3-to'plam	1	3	2	5
	3	3	4	-8
	5	4	6	4
	7	3	8	3
4-to'plam	1	4	2	3
	3	-5	4	2
	5	4	6	3
	7	5	8	4
5-to'plam	1	9	2	14
	3	36	4	4
	5	4	6	25
	7	8	8	7
7-to'plam	1	11	2	5
	3	-2	4	5
	5	4	–	–
6-to'plam	1. 5 faqat o'ng tomonga qo'shilgan – ikkala tomonga qo'shish kerak. To'g'ri: $x = 12 + 5 = 17$.			
	2. -5 ishorasi almashtirilmagan. To'g'ri: $2x = 9 + 5 = 14$, $x = 7$.			
	3. -2 faqat x ga taqsimlangan, -3 ga emas. To'g'ri: $-2x + 6 = 10 \Rightarrow x = -2$.			
	4. $3x$ va 5 o'xshash had emas. To'g'ri: $3x = 15 \Rightarrow x = 5$.			
	5. $x = 5$ to'g'ri, lekin savol $3x$ ni so'ragan. To'g'ri: $3x = 15$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Bir xil asosli daraja qoidalari

Qoida	Misol
$a^m \times a^n = a^{m+n}$	$2^3 \times 2^4 = 2^{3+4} = 2^7$
$(a^m)^n = a^{mn}$	$(2^7)^2 = 2^{7\times 2} = 2^{14}$
$\frac{a^m}{a^n} = a^{m-n}$	$\frac{2^{14}}{2^5} = 2^{14-5} = 2^9$

QOIDA · Nol va manfiy daraja

Nolmas a uchun: $a^0 = 1$. Manfiy ko'rsatkich – teskari sonni bildiradi, asosni manfiy qilmaydi:

$$5^0 + 2^{-3} \times 4^2$$

← 1. $a^0=1$, $a^{-n}=\frac{1}{a^n}$

$$\downarrow$$

$$1 + \frac{1}{8} \times 16$$

← 2. ko'paytiring

$$\downarrow$$

$$1 + 2 \qquad = 3$$

QOIDA · Kvadrat ildizni soddalashtirish

Eng katta to'liq kvadrat ko'paytuvchini toping, so'ng ajrating: $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$.

$$\sqrt{72}$$

← 1. $72=36\times 2$, 36 – to'liq kvadrat

$$\downarrow$$

$$\sqrt{36 \times 2} = \sqrt{36} \times \sqrt{2}$$

← 2. ajrating

$$\downarrow$$

$$= 6\sqrt{2}$$

Eslatma: $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$ *– masalan* $\sqrt{16+9} = 5$, *biroq* $\sqrt{16} + \sqrt{9} = 7$.

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
x^2	exponent / power	*daraja*
a (in a^n)	base	*asos*
n (in a^n)	exponent (index)	*ko'rsatkich*
a^0	zero exponent	*nol darajasi*
a^{-n}	negative exponent	*manfiy daraja*
$\frac{1}{a}$	reciprocal	*teskari son*
$\sqrt{a}$	square root	*kvadrat ildiz*
$36 = 6^2$	perfect square	*to'liq kvadrat*
–	value	*qiymat*
=	equal	*teng*
≡	equivalent	*teng kuchli*
+	positive	*musbat*
–	negative	*manfiy*
–	real	*haqiqiy*
$\mathbb{Z}$	integer	*butun son*
$\frac{a}{b}$	fraction	*kasr*
–	solve	*yechmoq*
–	solution	*yechim*
–	equation	*tenglama*
–	expression	*ifoda*
–	simplifying	*soddalashtirish*
–	error	*xato*
–	factor	*ko'paytuvchi*
x	variable	*o'zgaruvchi*
x^2	square	*kvadrat*
$\sqrt{}$	root	*ildiz*

1-to'plam. *Daraja qoidalari – nol/manfiy ko'rsatkichsiz sof mashq.*

1. $2^2 \times 2^3 =$

2. $\frac{3^4}{3^2} =$

3. $(2^2)^3 =$

4. $5^3 \times 5^2 =$

5. $\frac{4^5}{4^3} =$

6. $(3^2)^2 =$

7. $2^3 \times 2^3 =$

8. $(2^3)^3 =$

2-to'plam. *Nol va manfiy ko'rsatkichlar.*

1. $5^0 =$

2. $2^{-3} =$

3. $4^{-2} =$

4. $2^0 \times 2^{-2} =$

5. $3^{-2} \times 3^4 =$

6. $5^2 \times 5^{-2} =$

7. $10^{-1} \times 100 =$

8. $2^{-1} + 2^{-2} =$

3-to'plam. *Aralash – haqiqiy test zichligi.*

1. $\dfrac{(2^2 \times 2^3)^2}{2^4} =$

2. $(3 \times 3^2)^2 =$

3. $\dfrac{5^3}{5^2 \times 5^{-1}} =$

4. $(2^{-1})^{-3} =$

5. $(4^2)^0 =$

6. $\dfrac{2^5}{2^{-2}} =$

7. $\dfrac{(3^2)^2}{3^3} =$

8. $10^3 \times 10^{-1} \div 10 =$

4-to'plam. *Kvadrat ildizlarni soddalashtirish.*

1. $\sqrt{16} =$

2. $\sqrt{49} =$

3. $\sqrt{18} =$

4. $\sqrt{50} =$

5. $\sqrt{72} =$

6. $\sqrt{20} =$

7. $\sqrt{45} =$

8. $\sqrt{8} =$

5-to'plam. *Ildizlarni qo'shish va ayirish.*

1. $\sqrt{8} + \sqrt{2} =$

2. $\sqrt{18} - \sqrt{2} =$

3. $\sqrt{12} + \sqrt{27} =$

4. $\sqrt{50} - \sqrt{8} =$

5. $2\sqrt{3} \times \sqrt{3} =$

6. $\sqrt{2} \times \sqrt{8} =$

7. $3\sqrt{5} + 2\sqrt{5} =$

8. $\sqrt{45} - \sqrt{20} =$

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. $2^3 \times 2^3$ ifodasida ko'rsatilgan yechimda xato bor:
   "$2^3 \times 2^3 = 2^9 = 512$."
   Xato va to'g'ri javob: .....

2. $(2^2)^3$ ifodasida ko'rsatilgan yechimda xato bor:
   "$(2^2)^3 = 2^{2+3} = 2^5 = 32$."
   Xato va to'g'ri javob: .....

3. 2^{-3} ifodasida ko'rsatilgan yechimda xato bor:
   "$2^{-3} = -8$."
   Xato va to'g'ri javob: .....

4. $\sqrt{16 + 9}$ ifodasida ko'rsatilgan yechimda xato bor:
   "$\sqrt{16} + \sqrt{9} = 4 + 3 = 7$."
   Xato va to'g'ri javob: .....

5. $\sqrt{32}$ ifodasida ko'rsatilgan yechimda xato bor:
   "$32 = 16 \times 2$ *deb to'g'ri ajratilgan, lekin javobda* $16\sqrt{2}$ *deb yozilgan.*"
   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $\dfrac{(2^3)^2}{2^4} + \sqrt{16} =$

2. $\sqrt{3^2 \times 2^2} =$

3. $2^{-2} \times 4^2 + \sqrt{9} =$

4. $(\sqrt{2})^4 =$

5. $1 + \sqrt{5^2 - 3^2} =$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	32	2	9
	3	64	4	3125
	5	16	6	81
	7	64	8	512
2-to'plam	1	1	2	$\frac{1}{8}$
	3	$\frac{1}{16}$	4	$\frac{1}{4}$
	5	9	6	1
	7	10	8	$\frac{3}{4}$
3-to'plam	1	64	2	729
	3	25	4	8
	5	1	6	128
	7	3	8	10
4-to'plam	1	4	2	7
	3	$3\sqrt{2}$	4	$5\sqrt{2}$
	5	$6\sqrt{2}$	6	$2\sqrt{5}$
	7	$3\sqrt{5}$	8	$2\sqrt{2}$
5-to'plam	1	$3\sqrt{2}$	2	$2\sqrt{2}$
	3	$5\sqrt{3}$	4	$3\sqrt{2}$
	5	6	6	4
	7	$5\sqrt{5}$	8	$\sqrt{5}$
7-to'plam	1	8	2	6
	3	7	4	4
	5	5	–	–
6-to'plam	1. Ko'rsatkichlarni ko'paytirgan, qo'shish kerak edi (ko'paytma qoidasi). To'g'ri: $2^{3+3} = 2^6 = 64$.			
	2. Ko'rsatkichlarni qo'shgan, ko'paytirish kerak edi (daraja darajasi). To'g'ri: $2^{2\times3} = 2^6 = 64$.			
	3. Manfiy ko'rsatkichni asosni manfiy qiladi deb tushungan. To'g'ri: $2^{-3} = \frac{1}{8}$.			
	4. Yig'indining ildizini ildizlar yig'indisiga bo'lgan – bu hech qachon to'g'ri emas. To'g'ri: $\sqrt{25} = 5$.			
	5. 16 ni to'g'ri ajratgan, lekin uning ildizini olmagan. To'g'ri: $\sqrt{16 \times 2} = 4\sqrt{2}$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

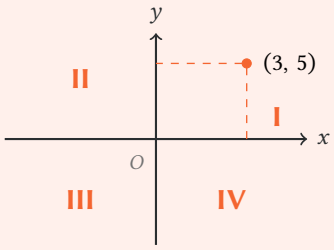
YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Koordinatalar tekisligi



Chorak	x	y
I	+	+
II	−	+
III	−	−
IV	+	−

QOIDA · $y = kx + b$

$k = -3$ **(qiyalik/slope)**

$$y = -3x + 5$$

$b = 5$ **(Oy-kesma)**

SAT'da k o'rniga m harfi va "slope" so'zi ishlatiladi: $y = mx + b$. $k = m$ – faqat nom boshqa.

QOIDA · $f(x)$ yozuvi

$f(x)$ – bu shunchaki y ning boshqa nomi. x o'rniga son qo'yib, ifodani hisoblang:

$$f(x) = 2x + 3$$
$$f(4) = 2(4) + 3$$
$$= 8 + 3$$
$$f(4) = 11$$

$f(x) = x^2 - 2x$

← manfiy son qavs ichida!

$f(-3) = \underline{(-3)}^2 - 2\underline{(-3)}$

$= 9 + 6 = 15$

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
(x, y)	**point**	*nuqta*
–	**points**	*nuqtalar*
–	**coordinate**	*koordinata*
(x, y)	**coordinates**	*koordinatalar*
–	**plane**	*tekislik*
–	**axis**	*o'q*
$(0, 0)$	**origin**	*koordinata boshi*
–	**quadrant**	*chorak (kvadrant)*
–	**linear**	*chiziqli*
$f(x)$	**function**	*funksiya*
–	**functions**	*funksiyalar*
–	**value**	*qiymat*
–	**values**	*qiymatlar*
–	**negative**	*manfiy*
+	**positive**	*musbat*
–	**equation**	*tenglama*
–	**equations**	*tenglamalar*
m	**slope**	*qiyalik*
b	**intercept**	*kesma*
–	**graph**	*grafik*
–	**line**	*chiziq*
–	**given**	*berilgan*
–	**represents**	*ifodalaydi*
–	**substituting**	*o'rniga qo'yish*
–	**constant**	*doimiy*
a	**coefficient**	*koeffitsient*
–	**horizontal**	*gorizontal*
–	**vertical**	*vertikal*
x^2	**squared**	*kvadratga ko'tarilgan*

1-to'plam. *Nuqta qaysi chorakka tegishli ekanini aniqlang.*

1. $(3,\ 5)$ → chorak:

2. $(-2,\ 4)$ → chorak:

3. $(-6,\ -1)$ → chorak:

4. $(7,\ -3)$ → chorak:

5. $(-4,\ -8)$ → chorak:

6. $(5,\ -5)$ → chorak:

7. $(-9,\ 2)$ → chorak:

8. $(1,\ 1)$ → chorak:

2-to'plam. *Tenglamadan k (burchak koeffitsienti) va b ni ajrating.*

1. $y = 3x + 7$ → $k=$ $b=$

2. $y = -2x + 1$ → $k=$ $b=$

3. $y = x - 5$ → $k=$ $b=$

4. $y = -x + 9$ → $k=$ $b=$

5. $y = 4x$ → $k=$ $b=$

6. $y = -6$ → $k=$ $b=$

7. $y = 0.5x - 3$ → $k=$ $b=$

8. $y = \frac{1}{3}x - 2$ → $k=$ $b=$

3-to'plam. *$f(a)$ ni musbat qiymatlar uchun hisoblang.*

1. $f(x) = 2x + 3.\ f(4) =$

2. $f(x) = 5x - 1.\ f(2) =$

3. $f(x) = -3x + 4.\ f(1) =$

4. $f(x) = x + 10.\ f(0) =$

5. $f(x) = 4x.\ f(3) =$

6. $f(x) = -x + 6.\ f(5) =$

7. $f(x) = 2x - 7.\ f(10) =$

8. $f(x) = -5x + 2.\ f(3) =$

4-to'plam. *$f(a)$ ni manfiy qiymatlar uchun hisoblang – manfiy son har doim qavsda.*

1. $f(x) = 3x + 2.\ f(-2) =$

2. $f(x) = -4x + 1.\ f(-3) =$

3. $f(x) = x + 8.\ f(-8) =$

4. $f(x) = x^2.\ f(-3) =$

5. $f(x) = x^2 + 1.\ f(-4) =$

6. $f(x) = 2x^2 - 3.\ f(-2) =$

7. $f(x) = -x^2.\ f(-5) =$

8. $f(x) = (x + 1)^2.\ f(-6) =$

5-to'plam. *Aralash – shaklni almashtiring, qiymatni hisoblang, tenglamani yeching.*

1 $y = 5x - 3$ ni $f(x)$ shaklida yozing va $f(2)$ ni toping.

2 $f(x) = -2x + 8$ ni $y = kx + b$ shaklida yozing, k, b ni toping.

3 $y = x + 1$ ni $f(x)$ shaklida yozing, so'ng $f(-4)$ ni hisoblang.

4 $f(x) = 4x - 9$. Agar $f(x) = 11$ bo'lsa, x ni toping.

5 $f(x) = -3x + 6$. Agar $f(x) = 0$ bo'lsa, x ni toping.

6 $y = -x - 5$ ni $f(x)$ shaklida yozing va $f(0)$ ni toping.

7 $f(x) = 2x + 1$ bo'lsa, $f(3) + f(-3)$ ni hisoblang.

8 $f(x) = x^2 - 4$. Agar $f(a) = 5$ bo'lsa, a ning musbat qiymatini toping.

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1 $f(x) = 3x + 2$. $f(5)$ ni toping:
"$f(5) = f \cdot 5$, f *noma'lum, javob topib bo'lmaydi.*"
Xato va to'g'ri javob: .....

2 $f(x) = x^2$. $f(-4)$ ni toping:
"$f(-4) = -4^2 = -16$."
Xato va to'g'ri javob: .....

3 $(-3,\ 5)$ nuqtasi qaysi chorakda?
"$x = -3$ *ni musbat deb o'ylab, I chorak.*"
Xato va to'g'ri javob: .....

4 $f(x) = 2x - 5$. Agar $f(x) = 7$, x ni toping:
"$2x - 5 = 7 \Rightarrow 2x = 2 \Rightarrow x = 1$."
Xato va to'g'ri javob: .....

5 $f(x) = x^2 + 3$. $f(-2)$ ni toping:
"$f(-2) = -2^2 + 3 = -4 + 3 = -1$."
Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1 $f(x) = 2x^2 - 3x + 1$. $f(-2) =$

2 $f(x) = -(x-3)^2$. $f(6) =$

3 $y = -4x + 11$ ni $f(x)$ shaklida yozing, so'ng $f(x) = 3$ bo'lganda x ni toping.

4 $f(x) = x^2 - 2x$. Agar $f(a) = 8$ bo'lsa, a ning manfiy qiymatini toping.

5 $f(x) = 3x + c$. Agar $f(2) = 10$ bo'lsa, c ni toping.

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	I	2	II
	3	III	4	IV
	5	III	6	IV
	7	II	8	I
2-to'plam	1	k=3, b=7	2	k=−2, b=1
	3	k=1, b=−5	4	k=−1, b=9
	5	k=4, b=0	6	k=0, b=−6
	7	k=0.5, b=−3	8	$k=\frac{1}{3}$, b=−2
3-to'plam	1	11	2	9
	3	1	4	10
	5	12	6	1
	7	13	8	−13
4-to'plam	1	−4	2	13
	3	0	4	9
	5	17	6	5
	7	−25	8	25
5-to'plam	1. $f(x) = 5x-3$; $f(2) = 7$ 2. $y = -2x+8$; $k = -2$, $b = 8$			
	3. $f(x) = x + 1$; $f(-4) = -3$ 4. $x = 5$			
	5. $x = 2$ 6. $f(x) = -x - 5$; $f(0) = -5$			
	7. $f(3) + f(-3) = 7 + (-5) = 2$			
	8. $a = 3$			
7-to'plam	1	15	2	−9
	3. $f(x) = -4x + 11$; $x = 2$ 4. $a = -2$ 5. $c = 4$			
6-to'plam	1. f funksiya nomi, songa ko'paytirilmaydi. To'g'ri: $f(5) = 3(5) + 2 = 17$.			
	2. Manfiy kiritmada qavs tushib qolgan. To'g'ri: $f(-4) = (-4)^2 = 16$.			
	3. x ning ishorasi noto'g'ri o'qilgan. To'g'ri: $x = -3<0$, $y=5>0 \Rightarrow$ II chorak.			
	4. $7 + 5 = 12$, 2 emas. To'g'ri: $2x = 12 \Rightarrow x = 6$.			
	5. Yana qavs tushib qolgan. To'g'ri: $f(-2) = (-2)^2+3 = 7$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · 1–8-darslar – asosiy formulalar

#	Mavzu	Asosiy qoida
1	Amallar tartibi	Qavs → Daraja/Ildiz → ×÷ → +−
2	Kasrlar	$\frac{a}{b} \pm \frac{c}{b}$: umumiy maxraj; $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$
3	O'nli kasr / foiz	%→o'nli kasr: ÷100; o'nli kasr → %: ×100
4	Ishorali sonlar	Xil ishora: ayirib, kattaning ishorasi. (×, ÷): bir xil ishora → +, har xil → −
5	Ifodalar	$a(b + c) = ab + ac$; o'xshash hadlar: koeffitsientlar qo'shiladi
6	Tenglamalar	Ikkala tomonga bir xil teskari amalni qo'llang
7	Daraja / ildiz	$a^m \cdot a^n = a^{m+n}$; $\frac{a^m}{a^n} = a^{m-n}$; $a^0 = 1$
8	Koordinata / funksiya	$y = kx + b \equiv f(x) = kx + b$; qiyalik $= k$

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
=	**equation**	*tenglama*
=	**equations**	*tenglamalar*
–	**expression**	*ifoda*
–	**expressions**	*ifodalar*
x	**variable**	*o'zgaruvchi*
x, y	**variables**	*o'zgaruvchilar*
a	**constant**	*doimiy*
a, b	**constants**	*o'zgarmaslar*
–	**term**	*had*
–	**terms**	*hadlar*
$\frac{a}{b}$	**fraction**	*kasr*
0.5	**decimal**	*o'nlik*
a (in $\frac{a}{b}$)	**numerator**	*surat*
b (in $\frac{a}{b}$)	**denominator**	*maxraj*
%	**percent**	*foiz*
%	**percentage**	*foiz*
–	**negative**	*manfiy*
+	**positive**	*musbat*
x^2	**exponent**	*daraja*
x^n	**exponents**	*daraja*
$\sqrt{x}$	**root**	*ildiz*
$f(x)$	**function**	*funksiya*
$f(x), g(x)$	**functions**	*funksiyalar*
–	**solve**	*yechmoq*
–	**solving**	*yechish*
–	**solution**	*yechim*
–	**solutions**	*yechimlar*
–	**value**	*qiymat*
–	**order**	*tartib*
–	**error**	*xato*

1-to'plam. *Amallar tartibi va kasrlarni birlashtiring.*

1. $\frac{3}{4} + 2 \times \left(\frac{1}{2} - \frac{1}{8}\right) =$

2. $\left(\frac{2}{3} + \frac{1}{6}\right) \times 4 - 1 =$

3. $10 - 3 \times \left(\frac{2}{5} + \frac{1}{10}\right) =$

4. $\left(\frac{7}{8} - \frac{1}{4}\right) \div \frac{1}{2} =$

5. $2 \times \frac{3}{4} + \left(1 - \frac{1}{3}\right) \times 6 =$

6. $\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{6}\right) \times 12 - 5 =$

2-to'plam. *Ishorali sonlar va darajalarni birlashtiring.*

1. $(-2)^3 - 3 \times (-4) =$

2. $-3^2 + (-3)^2 =$

3. $(-5)^2 \div (-5) + 2^3 =$

4. $-2 \times (-3)^2 + (-4) =$

5. $(-1)^5 \times (-2)^3 =$

6. $4 - (-2)^4 \div 2 =$

3-to'plam. *O'nli kasr/foiz va tenglamalarni birlashtiring. x ni toping.*

1. $0.25x + 10 = 35$

2. 40% of x is 32

3. $x - 15\%$ of $x = 68$

4. $0.5(x + 4) = 13$

5. 20% of $(x + 10) = 16$

6. $x + 0.1x = 44$

4-to'plam. *Algebraik ifodalar va tenglamalarni birlashtiring. x ni toping.*

1. $3(x - 2) + 5 = 2x + 7$

2. $2(x + 3) - (x - 1) = 10$

3. $4x - (2x + 5) = 7$

4. $5(x - 1) = 3(x + 3)$

5. $-2(x + 4) + 6 = x - 2$

6. $3(2x - 1) - 4 = 2(x + 5)$

5-to'plam. *Kasrli tenglamalarni yeching. x ni toping.*

1. $\frac{2}{3}x - 1 = 5$

2. $\frac{x}{4} + 3 = 7$

3. $\frac{3}{4}x + \frac{1}{2} = 2$

4. $\frac{x+2}{3} = 5$

5. $\frac{2x-1}{5} = 3$

6. $\frac{x}{2} - \frac{x}{3} = 2$

6-to'plam. *$f(x)$ funksiyasi, ishorali sonlar va darajalar.*

1. $f(x) = -2x^2 + 3$. $f(-2) =$

2. $f(x) = x^3 - 4x$. $f(-1) =$

3. $f(x) = (x-2)^2$. $f(-3) =$

4. $f(x) = 2x^2 - 5x + 1$. $f(-1) =$

5. $f(x) = -x^2 + 2x$. $f(3) =$? $f(-3) =$

6. $f(x) = kx + b$. $f(2) = 7, f(0) = 3$. $k =$? $b =$

7-to'plam. *Xatoni toping – ikki mavzu aralashgan, bitta xato qadam bor.*

1 $\frac{1}{2}x + 3 = 7$ ifodasida ko'rsatilgan yechimda xato bor:

"$\frac{1}{2}x = 7 + 3 = 10$, *shuning uchun* $x = 20$."

Xato va to'g'ri javob: .....

2 $3(x - 2) = 9$ ifodasida ko'rsatilgan yechimda xato bor:

"$3x - 2 = 9$, *shuning uchun* $3x = 11$, $x = \frac{11}{3}$."

Xato va to'g'ri javob: .....

3 $f(x) = x^2 - 2x$, $f(-3)$ ifodasida ko'rsatilgan yechimda xato bor:

"$f(-3) = (-3)^2 - 2(-3) = 9 - 6 = 3$."

Xato va to'g'ri javob: .....

4 $0.4x = 24$ ifodasida ko'rsatilgan yechimda xato bor:

"$x = 24 \times 0.4 = 9.6$."

Xato va to'g'ri javob: .....

5 $2(x + 1) - x = 9$ ifodasida ko'rsatilgan yechimda xato bor:

"$2x + 1 - x = 9$, $x + 1 = 9$, $x = 8$."

Xato va to'g'ri javob: .....

8-to'plam. *Bonus / eng qiyin – bitta masalada 3+ ko'nikma.*

1 $\frac{1}{2}(x - 4) + 0.25x = 5$

2 $f(x) = -2(x - 1)^2 + 3$. $f(-1) =$

3 30% of $(2x + 10) = x + 2$

4 $\frac{3}{4}x - (-2)^3 = 11$

5 $f(x) = kx + b$. $f(-2) = 1$, $f(2) = 9$. $k =$? $b =$? $f(0) =$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$\frac{3}{2}$	2	$\frac{7}{3}$
	3	$\frac{17}{2}$	4	$\frac{5}{4}$
	5	$\frac{11}{2}$	6	7
2-to'plam	1	4	2	0
	3	3	4	-22
	5	8	6	-4
3-to'plam	1	$x=100$	2	$x=80$
	3	$x=80$	4	$x=22$
	5	$x=70$	6	$x=40$
4-to'plam	1	$x=8$	2	$x=3$
	3	$x=6$	4	$x=7$
	5	$x=0$	6	$x=\frac{17}{4}$
5-to'plam	1	$x=9$	2	$x=16$
	3	$x=2$	4	$x=13$
	5	$x=8$	6	$x=12$
6-to'plam	1	-5	2	3
	3	25	4	8
	5	$f(3)=-3$, $f(-3)=-15$		
	6	$k=2$, $b=3$		
7-to'plam	1. Teskari amal xato – 3 qo'shish o'rniga ayirish kerak edi. To'g'ri: $\frac{1}{2}x = 4$, $x = 8$.			
	2. 3 faqat x ga taqsimlangan, -2 ga emas. To'g'ri: $3x - 6 = 9$, $x = 5$.			
	3. $-2 \times (-3) = -6$ emas, $+6$. To'g'ri: $9 + 6 = 15$.			
	4. Ko'paytirish o'rniga bo'lish kerak edi. To'g'ri: $x = 24 \div 0.4 = 60$.			
	5. 2 faqat x ga taqsimlangan, $+1$ ga emas. To'g'ri: $2x + 2 - x = 9$, $x = 7$.			
8-to'plam	1	$x=\frac{28}{3}$		
	2	-5	3	$x=\frac{5}{2}$
	4	$x=4$	5	$k=2, b=5$

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
about	*taxminan*
absolute	*mutlaq*
according	*ga ko’ra*
account	*hisob*
activity	*faoliyat*
actual	*haqiqiy*
acute	*o’tkir*
added	*qo’shilgan*
adding	*qo’shish*
additional	*qo’shimcha*
additionally	*bundan tashqari*
adjacent	*qo’shni*
adult	*katta yoshli*
adults	*kattalar*
advanced	*ilg’or*
against	*qarshi*
age	*yosh*
air	*havo*
algebra	*algebra*
already	*allaqachon*
alternate	*almashinuvchi*
altitude	*balandlik*
amount	*miqdor*
analysis	*tahlil*
angle	*burchak*

English	O’zbekcha
angles	*burchaklar*
animal	*hayvon*
animals	*hayvonlar*
annual	*yillik*
answer	*javob*
appears	*ko’rinadi*
applied	*qo’llanildi*
applying	*qo’llash*
approach	*yondashuv*
appropriate	*mos*
approximate	*taxminiy*
approximately	*taxminan*
april	*aprel*
arc	*yoy*
area	*yuza*
areas	*yuzalar*
article	*maqola*
ascending	*o’suvchi (tartib)*
asked	*so’ralgan*
assigned	*tayinlangan*
associated	*bog’liq*
assuming	*faraz qilib*
attended	*qatnashgan*
attendee	*ishtirokchi*
attendees	*ishtirokchilar*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
auger	*parma*
august	*avgust*
average	*o’rtacha*
away	*uzoqlikda*
axis	*o’q*
azaleas	*azaliya*
bacteria	*bakteriyalar*
bag	*xalta*
bags	*sumkalar*
balance	*balans*
ball	*to’p*
bank	*bank*
bar	*ustun*
base	*asos*
based	*asoslangan*
batch	*partiya*
beach	*plyaj*
beads	*munchoqlar*
beetle	*qo’ng’iz*
began	*boshlamoq*
best	*eng yaxshi*
bicycle	*velosiped*
biked	*velosipedda yurdi*
biking	*velosiped haydash*
bill	*hisob (to’lov)*

English	O’zbekcha
bin	*quti*
birch	*qayin*
bird	*qush*
black	*qora*
blackberries	*yejevika*
blood	*qon*
blue	*ko’k*
board	*taxta*
boards	*taxtalar*
boat	*qayiq*
body	*tana*
boiling	*qaynash*
book	*kitob*
books	*kitoblar*
born	*tug’ilgan*
bottles	*shishalar*
bottom	*tag*
bought	*sotib oldi*
boundary	*chegara*
bowl	*kosa*
box	*quti*
boxes	*qutilar*
boxwoods	*buksus*
brakes	*tormozlar*
budget	*byudjet*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
buildings	*binolar*
bursts	*yoriladi*
bus	*avtobus*
bushels	*bushel*
business	*biznes*
button	*tugma*
buttons	*tugmalar*
buy	*sotib olmoq*
buys	*sotib oladi*
calcium	*kaltsiy*
calculated	*hisoblangan*
calculating	*hisoblash*
calculation	*hisoblash*
calories	*kaloriyalar*
camping	*kemping*
candle	*sham*
candles	*shamlar*
cannot	*mumkin emas*
capacity	*sig’im*
car	*avtomobil*
card	*karta*
cards	*kartalar*
cargo	*yuk*
cars	*mashina*
cell	*katak*

English	O’zbekcha
cells	*katakcha*
celsius	*selsiy*
center	*markaz*
centimeter	*santimetr*
centimeters	*santimetr*
cereal	*donli taom*
certain	*ma’lum (bir)*
change	*o’zgarish*
charge	*to’lov*
charged	*to’lov olingan*
charges	*to’lovlar*
cheese	*pishloq*
child	*bola*
children	*bolalar*
chips	*chipslar*
chocolate	*shokolad*
chose	*tanladi*
chosen	*tanlangan*
circle	*doira*
circular	*doiraviy*
circumference	*aylana uzunligi*
city	*shahar*
class	*sinf*
classes	*toifalar*
clients	*mijozlar*

VOCABULARY – COMPLETE GLOSSARY

English	O'zbekcha
closest	*eng yaqin*
club	*to'garak*
cluster	*klaster*
coefficient	*koeffitsient*
coefficients	*koeffitsientlar*
coin	*tanga*
coins	*tangalar*
collect	*to'plamoq*
collected	*to'plandi*
collection	*to'plam*
colony	*koloniya*
color	*rang*
colored	*rangli*
combinations	*kombinatsiyalar*
combined	*qo'shilgan*
combining	*birlashtirish*
comic	*komiks*
comments	*sharhlar*
common	*umumiy*
company	*kompaniya*
compare	*solishtirmoq*
compares	*solishtiradi*
complete	*to'ldirmoq*
completed	*bajarilgan*
computational	*hisoblash*

English	O'zbekcha
computer	*kompyuter*
conceptual	*tushunchaviy*
conclusion	*xulosa*
concrete	*beton*
condition	*shart*
conditional	*shartli*
cone	*konus*
congruent	*teng (kongruent)*
consecutive	*ketma-ket*
consisting	*...dan iborat*
consists	*iborat*
constant	*doimiy*
constants	*o'zgarmaslar*
constructed	*qurilgan*
contain	*o'z ichiga olmoq*
container	*idish*
containers	*idishlar*
containing	*o'z ichiga olgan*
contains	*o'z ichiga oladi*
context	*kontekst*
cookies	*pechenye*
coordinate	*koordinata*
coordinates	*koordinatalar*
corn	*makkajo'xori*
correctly	*to'g'ri*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
correspond	*mos kelmoq*
corresponding	*mos*
corresponds	*mos keladi*
cosine	*kosinus*
cost	*narx*
costs	*turadi (narxi ... bo’ladi)*
country	*mamlakat*
courses	*kurs*
crackers	*krekerlar*
crates	*yashiklar*
cream	*qaymoq*
create	*yaratmoq*
created	*yaratilgan*
crosses	*kesib o’tadi*
cube	*kub*
cubes	*kub*
cubic	*kubik*
cup	*stakan*
cups	*stakanlar*
customer	*mijoz*
customers	*mijozlar*
cutting	*kesish*
cylinder	*silindr*
daily	*kunlik*
dance	*raqs*

English	O’zbekcha
data	*ma’lumotlar*
day	*kun*
daylight	*kun yorug’ligi*
days	*kunlar*
dealer	*sotuvchi*
decay	*kamayish*
decimal	*o’nlik*
decrease	*kamayish*
decreased	*kamaymoq*
decreases	*kamayadi*
decreasing	*kamayuvchi*
defective	*nuqsonli*
define	*ta’riflamoq*
defined	*aniqlangan*
defines	*aniqlaydi*
defining	*aniqlovchi*
definition	*ta’rif*
degree	*gradus*
degrees	*gradus*
deliveries	*yetkazib berishlar*
denominator	*maxraj*
density	*zichlik*
department	*bo’lim*
deposit	*depozit*
deposits	*depozitlar*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
depth	*chuqurlik*
descending	*kamayish tartibida*
describes	*tasvirlamoq*
designer	*dizayner*
determine	*aniqlamoq*
determined	*aniqlangan*
determining	*aniqlash*
deviation	*chetlanish*
device	*qurilma*
diagonal	*diagonal*
diameter	*diametr*
difference	*ayirma*
different	*turli*
discount	*chegirma*
discounted	*chegirma qilingan*
discriminant	*diskriminant*
dispenser	*tarqatgich*
displays	*ko’rsatadi*
distance	*masofa*
distinct	*farqli*
distributing	*qavsni ochish*
distribution	*taqsimot*
distributions	*taqsimotlar*
distributive	*taqsimot*
diver	*sho’ng’uvchi*

English	O’zbekcha
divided	*bo’lingan*
dividing	*bo’lish*
doesn’t	*...emas (qilmaydi)*
dollars	*dollar*
dot	*nuqta*
double	*ikki baravar*
doubles	*ikki baravar ko’payadi*
dragonflies	*ninachilar*
drawing	*chizma*
driven	*haydalgan*
dropped	*tashlangan*
drove	*haydadi*
earn	*ishlab topmoq*
earned	*ishlab topdi*
earnings	*daromad*
earns	*ishlab topadi*
edge	*qirra*
egg	*tuxum*
eggs	*tuxumlar*
eighth	*sakkizinchi*
electric	*elektr*
elements	*elementlar*
elevation	*balandlik*
employee	*xodim*
employees	*xodimlar*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
end	*oxiri*
energy	*energiya*
enrolled	*ro’yxatdan o’tmoq*
enter	*kiritish*
entire	*butun*
equal	*teng*
equation	*tenglama*
equations	*tenglamalar*
equilateral	*teng tomonli*
equipment	*jihozlar*
equivalent	*teng kuchli*
error	*xato*
errors	*xatolar*
essay	*insho*
estimate	*baholamoq (taxmin qilish)*
estimated	*taxmin qilingan*
estimates	*baholaydi*
evaluating	*hisoblash*
even	*juft*
event	*hodisa*
exactly	*aynan*
example	*misol*
examples	*misollar*
exerted	*qo’llangan (kuch)*
expanding	*yoyish*

English	O’zbekcha
expected	*kutilgan*
expenses	*xarajatlar*
experiment	*tajriba*
exponent	*daraja*
exponential	*ko’rsatkichli*
exponents	*daraja*
express	*ifodalamoq*
expressed	*ifodalangan*
expresses	*ifodalaydi*
expression	*ifoda*
expressions	*ifodalar*
exterior	*tashqi*
fabric	*mato*
face	*yoq*
faces	*yoqlar*
factor	*ko’paytuvchi*
factored	*ko’paytuvchilarga ajratilgan*
factoring	*ko’paytuvchilarga ajratish*
factors	*ko’paytuvchilar*
fahrenheit	*farengeyt*
false	*noto’g’ri*
fee	*to’lov*
feet	*fut*
female	*ayol*
fertilizer	*o’g’it*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
fewer	*kamroq*
field	*maydon*
figure	*shakl*
filled	*to’ldirilgan*
filler	*to’ldiruvchi*
film	*film*
find	*topish*
finding	*topish*
first	*birinchi*
fish	*baliq*
fit	*mos kelish*
fixed	*o’zgarmas*
flags	*bayroqlar*
flight	*parvoz*
floor	*pol*
fluid	*suyuqlik*
fly	*uchmoq*
following	*quyidagi*
follows	*kelib chiqadi*
food	*oziq-ovqat*
foot	*fut*
forest	*o’rmon*
form	*ko’rinish*
formed	*hosil bo’lgan*
formula	*formula*

English	O’zbekcha
found	*topilgan*
fourth	*to’rtinchi*
fraction	*kasr*
free	*bepul*
frequency	*chastota*
fruit	*meva*
frying	*qovurish*
full	*to’liq*
function	*funksiya*
functions	*funksiyalar*
gallon	*gallon*
gallons	*gallonlar*
game	*o’yin*
games	*o’yinlar*
garden	*bog’*
gas	*gaz*
gasoline	*benzin*
geologist	*geolog*
geometry	*geometriya*
get	*olmoq*
gift	*sovg’a*
give	*bermoq*
given	*berilgan*
gives	*beradi*
goal	*maqsad*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
grade	*sinf*
gram	*gramm*
grams	*gramm*
graph	*grafik*
graphed	*grafikka tushirilgan*
graphs	*grafiklar*
gray	*kulrang*
greater	*katta*
greatest	*eng katta*
green	*yashil*
ground	*yer*
group	*guruh*
growth	*o’sish*
gym	*sportzal*
habitat	*yashash muhiti*
half	*yarim*
halfway	*yarim yo’lda*
hand	*qo’l*
haul	*tashimoq*
hectare	*gektar*
height	*balandlik*
heights	*balandliklar*
high	*yuqori*
highest	*eng yuqori*
histogram	*gistogramma*

English	O’zbekcha
hits	*tegadi (uriladi)*
home	*uy*
horizontal	*gorizontal*
hour	*soat*
hourly	*soatlik*
hours	*soat*
house	*uy*
houses	*uylar*
however	*ammo*
hypotenuse	*gipotenuza*
ice	*muz*
iceland	*Islandiya*
identical	*aynan bir xil*
igneous	*otqin*
inch	*dyuym*
inches	*dyuym*
include	*o’z ichiga olmoq*
included	*kiritilgan*
includes	*o’z ichiga oladi*
incorrectly	*noto’g’ri*
increase	*oshish*
increased	*ortdi*
increases	*ortadi*
increasing	*o’suvchi*
indicates	*ko’rsatadi*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
inequalities	*tengsizliklar*
inequality	*tengsizlik*
infinitely	*cheksiz*
information	*ma’lumot*
initial	*boshlang’ich*
inside	*ichida*
installation	*o’rnatish*
instead	*o’rniga*
integer	*butun son*
integers	*butun sonlar*
intercept	*kesma*
intercepts	*kesishish nuqtalari*
interior	*ichki*
intermediate	*oraliq*
internet	*internet*
internships	*stajirovka*
interpretation	*talqin*
intersect	*kesishmoq*
intersected	*kesishgan*
intersection	*kesishish*
intersects	*kesishadi*
interval	*oraliq*
isn’t	*emas*
isosceles	*teng yonli*
items	*buyumlar*

English	O’zbekcha
january	*yanvar*
jars	*bankalar*
job	*ish*
join	*qo’shilmoq*
joules	*joul*
juice	*sharbat*
july	*iyul*
kelvins	*kelvinlar*
kilograms	*kilogramm*
kilometer	*kilometr*
kilometers	*kilometr*
kinetic	*kinetik*
labeled	*belgilangan*
lake	*ko’l*
land	*yer*
large	*katta*
larger	*kattaroq*
largest	*eng katta*
last	*oxirgi*
lateral	*yon (yon sirt)*
launch	*uchirish*
launched	*uchirildi*
left	*chap*
leg	*katet*
legs	*katetlar*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
lemons	*limonlar*
length	*uzunlik*
lengths	*uzunliklar*
let	*faraz qilaylik*
level	*saviya*
lie	*yotmoq*
lies	*yotadi*
like	*o’xshash*
likely	*ehtimol*
line	*chiziq*
linear	*chiziqli*
lines	*chiziqlar*
liquid	*suyuqlik*
list	*ro’yxat*
listed	*ro’yxatda keltirilgan*
liter	*litr*
liters	*litr*
local	*mahalliy*
located	*joylashgan*
locations	*joylashuv*
long	*uzun*
longer	*uzunroq*
longest	*eng uzun*
low	*past*
lowest	*eng past*

English	O’zbekcha
machine	*mashina*
made	*qilingan*
magazines	*jurnallar*
maintain	*saqlab turish*
make	*yasamoq*
makes	*qilmoq*
making	*hosil qilish*
maple	*klyon*
marble	*sharcha*
marbles	*sharchalar*
margin	*chegara*
marked	*belgilamoq*
mass	*massa*
materials	*materiallar*
maximum	*maksimum*
mean	*o’rtacha*
means	*anglatadi*
measure	*o’lchov*
measured	*o’lchangan*
measures	*o’lchovlari*
measuring	*o’lchash*
median	*mediana*
medians	*medianalar*
meet	*uchrashmoq*
member	*a’zo*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
members	*a’zolar*
metal	*metall*
meter	*metr*
meters	*metr*
method	*usul*
middle	*o’rta*
midpoint	*o’rta nuqta*
mile	*milya*
miles	*milya*
milliliter	*millilitr*
millimeters	*millimetr*
million	*million*
minimum	*minimum*
minus	*minus*
minute	*daqiqa*
minutes	*daqiqa*
mirror	*ko’zgu*
misidentifying	*noto’g’ri aniqlash*
misinterpreting	*noto’g’ri talqin qilish*
mixture	*aralashma*
mode	*moda*
model	*model*
modeled	*modellashtirildi*
models	*modellar*
monarch	*monarx (kapalak turi)*

English	O’zbekcha
money	*pul*
month	*oy*
monthly	*oylik*
months	*oylar*
moon	*oy*
movie	*kino*
movies	*filmlar*
moving	*harakatlanayotgan*
muffins	*maffin*
multiplied	*ko’paytirildi*
multiply	*ko’paytirmoq*
multiplying	*ko’paytirish*
museum	*muzey*
nearest	*eng yaqin*
need	*kerak*
needed	*kerakli*
needs	*kerak bo’ladi*
negative	*manfiy*
never	*hech qachon*
new	*yangi*
next	*keyingi*
nonlinear	*nochiziqli*
normal	*normal*
note	*eslatma*
novels	*romanlar*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
number	*son*
numbers	*sonlar*
numerator	*surat*
object	*jism*
objects	*narsalar*
observation	*kuzatish*
obtuse	*o’tmas (burchak)*
occurs	*ro’y beradi*
ocean	*okean*
odd	*toq*
office	*ofis*
old	*eski*
onetime	*bir martalik*
online	*onlayn*
opened	*ochildi*
opens	*ochadi*
opposite	*qarama-qarshi*
oranges	*apelsinlar*
order	*tartib*
ordered	*tartiblangan*
original	*dastlabki*
ounces	*unsiya*
outcomes	*natijalar*
outfits	*kiyimlar*
output	*chiqish (natija)*

English	O’zbekcha
owner	*egasi*
packages	*paketlar*
paid	*to’lamoq*
paint	*bo’yoq*
pair	*juft*
pairs	*juftlar*
pans	*tovalar*
pants	*shim*
paper	*qog’oz*
parabola	*parabola*
parallel	*parallel*
parallelogram	*parallelogramm*
parentheses	*qavs*
park	*park*
parking	*avtoturargoh*
part	*qism*
participant	*ishtirokchi*
participants	*ishtirokchilar*
particular	*muayyan*
parts	*qismlar*
party	*ziyofat*
pass	*o’tmoq*
passes	*o’tmoq*
passing	*o’tish*
payment	*to’lov*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
pens	*ruchkalar*
people	*odamlar*
per	*har biriga*
percent	*foiz*
percentage	*foiz*
percentages	*foizlar*
perimeter	*perimetr*
perimeters	*perimetr*
period	*davr*
periods	*davrlar*
perpendicular	*perpendikulyar*
person	*odam*
phone	*telefon*
piece	*bo’lak*
pieces	*donalar*
placed	*joylashtirilgan*
plan	*reja*
plane	*tekislik*
planet	*sayyora*
plans	*rejalar*
plant	*o’simlik*
planted	*ekildi*
plants	*o’simliklar*
platform	*platforma*
plausible	*asosli*

English	O’zbekcha
play	*o’ynamoq*
played	*o’ynagan*
player	*o’yinchi*
players	*o’yinchilar*
plot	*belgilash*
plus	*plyus*
point	*nuqta*
points	*nuqtalar*
polynomial	*ko’phad*
population	*aholi*
positive	*musbat*
possible	*mumkin bo’lgan*
poster	*plakat*
posters	*plakatlar*
potatoes	*kartoshka*
pound	*funt*
pounds	*funt*
predicted	*bashorat qilingan*
predicts	*bashorat qiladi*
premium	*sug’urta badali*
prepare	*tayyorlash*
preserve	*saqlamoq*
pressure	*bosim*
previous	*oldingi*
price	*narx*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
prism	*prizma*
probability	*ehtimollik*
problem	*masala*
produce	*ishlab chiqarmoq*
produced	*ishlab chiqardi*
produces	*ishlab chiqaradi*
product	*ko’paytma*
production	*ishlab chiqarish*
profit	*foyda*
program	*dastur*
project	*loyiha*
projectile	*otilgan jism*
propellant	*yonilg’i*
properties	*xossalar*
property	*xossa*
proportion	*proportsiya*
proportional	*proporsional*
proposal	*taklif*
protein	*oqsil*
prove	*isbotlamoq*
purchase	*xarid*
purchased	*sotib olingan*
pyramid	*piramida*
pythagorean	*pifagor*
quadratic	*kvadrat*

English	O’zbekcha
quantity	*miqdor*
rabbit	*quyon*
radians	*radian*
radius	*radius*
ran	*yugurdi*
random	*tasodifiy*
randomly	*tasodifiy ravishda*
range	*diapazon*
raspberries	*malina*
rate	*tezlik*
rates	*tezliklar*
rather	*aksincha*
ratio	*nisbat*
ratios	*nisbatlar*
reach	*yetmoq*
reached	*yetdi*
reaches	*yetadi*
read	*o’qimoq*
real	*haqiqiy*
received	*oldi*
reciprocal	*teskari son/ifoda*
recorded	*qayd qilindi*
rectangle	*to’g’ri to’rtburchak*
rectangles	*to’g’ri to’rtburchaklar*
rectangular	*to’g’ri burchakli*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
red	*qizil*
reduced	*qisqartirilgan*
region	*soha*
registered	*ro’yxatdan o’tgan*
regular	*muntazam*
relates	*bog’liq*
relationship	*bog’liqlik*
relationships	*munosabatlar*
released	*chiqarilgan*
remain	*qolish*
remaining	*qolgan*
remains	*o’zgarmay qoladi*
removed	*olib tashlangan*
rent	*ijara haqi*
rental	*ijara*
renting	*ijaraga olish*
repair	*ta’mirlash*
reported	*qayd etilgan*
represent	*ifodalamoq*
represented	*ifodalangan*
representing	*ifodalovchi*
represents	*ifodalaydi*
residents	*aholi*
respectively	*mos ravishda*
responded	*javob berdi*

English	O’zbekcha
rest	*qolgan qism*
restaurant	*restoran*
restaurants	*restoranlar*
result	*natija*
resulting	*natijada hosil bo’lgan*
results	*natijalar*
retail	*chakana*
revenue	*daromad*
rewriting	*qayta yozish*
rewritten	*qayta yozilgan*
ride	*minmoq*
right	*to’g’ri*
rocket	*raketa*
rocks	*toshlar*
rolls	*nonchalar*
room	*xona*
root	*ildiz*
round	*yaxlitlamoq*
rows	*qatorlar*
running	*yugurish*
salads	*salatlar*
salary	*maosh*
sale	*sotuv*
sales	*sotuv*
salt	*tuz*

VOCABULARY – COMPLETE GLOSSARY

English	O'zbekcha
sample	*tanlanma*
samples	*namunalar*
sandwich	*sendvich*
satisfies	*qanoatlantiradi*
satisfy	*qanoatlantiradi*
save	*jamg'arish*
savings	*jamg'arma*
scale	*masshtab*
scatterplot	*sochilma diagramma*
scatterplots	*sochilish diagrammalari*
school	*maktab*
scientist	*olim*
screws	*vintlar*
scuba	*skuba*
sea	*dengiz*
seals	*tyulenlar*
seating	*o'tirish joylari*
seats	*o'rindiqlar*
second	*ikkinchi*
seconds	*soniya*
seeds	*urug'lar*
segment	*kesma*
segments	*kesmalar*
selected	*tanlangan*
selecting	*tanlash*

English	O'zbekcha
selling	*sotish*
sells	*sotmoq*
sent	*yubormoq*
september	*sentyabr*
service	*xizmat*
session	*mashg'ulot*
sessions	*mashg'ulotlar*
set	*to'plam*
sets	*to'plamlar*
setting	*o'rnatish*
shaded	*bo'yalgan*
shape	*shakl*
shifting	*siljish*
shipment	*yuk jo'natmasi*
shirt	*ko'ylak*
shirts	*ko'ylaklar*
shorter	*qisqaroq*
shortest	*eng qisqa*
show	*ko'rsatish*
shown	*ko'rsatilgan*
shows	*ko'rsatadi*
side	*tomon*
sides	*tomonlar*
sign	*ishora*
silicon	*kremniy*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
similar	*o’xshash*
similarly	*xuddi shunday*
simplifies	*soddalashtiradi*
simplifying	*soddalashtirish*
sine	*sinus*
site	*joy (maydon)*
situation	*vaziyat*
size	*o’lcham*
slope	*qiyalik*
slopes	*qiyaliklar*
small	*kichik*
smaller	*kichikroq*
smallest	*eng kichik*
snails	*shilliqqurtlar*
snow	*qor*
snowfall	*qor yog’ishi*
sold	*sotildi*
solution	*yechim*
solutions	*yechimlar*
solve	*yechmoq*
solved	*yechmoq*
solving	*yechish*
space	*bo’shliq*
species	*turlar*
speed	*tezlik*

English	O’zbekcha
spend	*sarflamoq*
spends	*sarflaydi*
spent	*sarflagan*
sphere	*shar*
spoons	*qoshiqlar*
sport	*sport*
spread	*yoyilish*
spring	*bahor*
square	*kvadrat*
squared	*kvadratga ko’tarilgan*
squares	*kvadratlar*
squaring	*kvadratga ko’tarish*
standard	*standart*
star	*yulduz*
stars	*yulduzlar*
start	*boshlanish*
started	*boshlandi*
starts	*boshlanadi*
state	*shtat*
statement	*tasdiq*
statements	*tasdiqlar*
states	*shtatlar*
station	*bekat*
statistics	*statistika*
storage	*saqlash*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
store	*do’kon*
straight	*to’g’ri*
student	*o’quvchi*
students	*o’quvchilar*
study	*tadqiqot*
subscribed	*obuna bo’ldi*
subscribers	*obunachilar*
substance	*modda*
substituted	*o’rniga qo’ymoq*
substituting	*o’rniga qo’yish*
substitution	*o’rniga qo’yish*
subtracting	*ayirish*
sufficient	*yetarli*
suits	*kostyumlar*
sum	*yig’indi*
summarizes	*umumlashtirmoq*
summer	*yoz*
supplementary	*to’ldiruvchi*
support	*qo’llab-quvvatlamoq*
surface	*sirt*
survey	*so’rovnoma*
surveyed	*so’rov o’tkazilgan*
swimmers	*suzuvchilar*
switching	*almashtirish*
symmetric	*simmetrik*

English	O’zbekcha
system	*sistema*
systems	*sistemalar*
table	*jadval*
tables	*jadvallar*
tablet	*planshet*
take	*olish*
takes	*oladi*
taking	*olish*
tall	*baland*
taller	*balandroq*
tangent	*urinma*
tank	*bak*
tape	*lenta*
tax	*soliq*
teacher	*o’qituvchi*
team	*jamoa*
teaspoons	*choy qoshiq*
teens	*o’smirlar*
televisions	*televizor*
temperature	*harorat*
temperatures	*haroratlar*
tent	*chodir*
tenth	*o’ninchi*
tents	*chodirlar*
term	*had*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
terms	*hadlar*
theater	*teatr*
theorem	*teorema*
therefore	*demak*
third	*uchinchi*
thousand	*ming*
through	*orqali*
thrown	*otilgan*
thus	*demak*
ticket	*chipta*
tickets	*chiptalar*
tiles	*plitka*
time	*vaqt*
times	*marta*
together	*birgalikda*
token	*jeton*
tokens	*jetonlar*
took	*olmoq*
total	*jami*
town	*shaharcha*
trail	*yo’lka*
trailer	*tirkama*
train	*poyezd*
training	*mashg’ulot*
translating	*tarjima qilish*

English	O’zbekcha
transversal	*kesuvchi*
trapezoid	*trapetsiya*
travel	*harakatlanish*
traveled	*bosib o’tdi*
traveling	*harakatlanmoq*
travels	*harakatlanadi*
tree	*daraxt*
trees	*daraxtlar*
triangle	*uchburchak*
triangles	*uchburchaklar*
trigonometry	*trigonometriya*
trip	*sayohat*
truck	*yuk mashinasi*
true	*to’g’ri*
twice	*ikki marta*
type	*tur*
types	*turlari*
unit	*birlik*
united	*birlashgan*
units	*birliklar*
upward	*yuqoriga*
use	*foydalanish*
used	*qo’llaniladi*
uses	*foydalanadi*
using	*yordamida*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
utility	*kommunal xizmat*
valley	*vodiy*
value	*qiymat*
values	*qiymatlar*
variable	*o’zgaruvchi*
variables	*o’zgaruvchilar*
vase	*vaza*
vehicle	*transport vositasi*
vertex	*cho’qqi*
vertical	*vertikal*
video	*video*
views	*ko’rishlar*
vine	*tok*
visitors	*tashrif buyuruvchilar*
volume	*hajm*
vote	*ovoz*
voted	*ovoz bermoq*
votes	*ovozlar*
wage	*ish haqi*
walk	*yurish*
walnuts	*yong’oq*
water	*suv*
wax	*mum*
ways	*usullar*
week	*hafta*

English	O’zbekcha
weighs	*tortadi (og’irligi)*
weight	*og’irlik*
weights	*og’irliklar*
whale	*kit*
white	*oq*
whole	*butun*
width	*kenglik*
within	*ichida*
won	*yutgan*
words	*so’zlar*
work	*ish*
worked	*ishlamoq*
worker	*ishchi*
working	*ishlash*
works	*ishlaydi*
worth	*qiymat*
written	*yozilgan*
yards	*yard*
year	*yil*
years	*yillar*
yellow	*sariq*
yield	*bermoq*
yields	*beradi*
zebras	*zebralar*
zero	*nol*

PHRASES – TEZ-TEZ UCHRAYDIGAN IBORALAR

Quyidagi iboralar College Board'ning rasmiy savollar bazasida eng ko'p uchraydigan, mavzudan qat'i nazar takrorlanadigan tayyor so'z birikmalaridir. Chiziqcha (...) o'rniga aniq son, o'zgaruvchi yoki ifoda qo'yiladi. Chastotasi bo'yicha tartiblangan – eng ko'p uchraydigani birinchi.

#	English	O'zbekcha
1	the value of	*...ning qiymati*
2	given that	*...berilgan bo'lsa*
3	the number of	*...soni*
4	both sides of	*...ning ikkala tomoni*
5	it follows that	*bundan kelib chiqadiki*
6	is equivalent to	*...ga teng (ekvivalent)*
7	the graph of	*...ning grafigi*
8	the length of	*...ning uzunligi*
9	represents the	*...ni ifodalaydi*
10	in the xy-plane	*xy-tekislikda*
11	the correct answer is	*to'g'ri javob – bu*
12	each side of	*har bir tomoni*
13	which of the following	*quyidagilardan qaysi biri*
14	dividing both sides	*ikkala tomonni bo'lish*
15	the measure of	*...ning o'lchami*
16	subtracting from both sides	*ikkala tomondan ayirish*
17	the y-intercept	*Y o'qidagi kesilgan nuqta*
18	the sum of	*...yig'indisi*
19	greater than	*...dan katta*
20	less than	*...dan kichik*
21	rather than	*...o'rniga*
22	the total number of	*jami soni*
23	there are	*mavjud*
24	can be represented by	*...bilan ifodalanishi mumkin*
25	in the form	*...ko'rinishida*
26	per hour	*soatiga*
27	at least	*kamida*
28	adding to both sides	*ikkala tomonga qo'shish*
29	more than	*...dan ortiq*
30	is approximately	*taxminan ...ga teng*

PHRASES – TEZ-TEZ UCHRAYDIGAN IBORALAR

#	English	O'zbekcha
31	passes through the point	*...nuqtadan o'tadi*
32	the slope of the line	*to'g'ri chiziqning qiyaligi*
33	corresponds to	*...ga mos keladi*
34	in terms of	*...orqali (ifodalangan)*
35	a solution to	*...ning yechimi*
36	the ratio of	*...nisbati*
37	according to the	*...ga ko'ra*
38	divided by	*...ga bo'lingan*
39	can be written as	*...shaklida yozilishi mumkin*
40	increased by	*...ga oshirilgan*
41	is represented by	*...bilan ifodalanadi*
42	is closest to	*...ga eng yaqin*
43	intersects the	*...ni kesib o'tadi*
44	results in	*natijasida ...kelib chiqadi*
45	based on the	*...asosida*
46	is given by	*...orqali beriladi*
47	the sum of the measures	*o'lchamlar yig'indisi*
48	for which	*qaysi ...uchun*
49	consists of	*...dan iborat*
50	instead of	*...o'rniga*
51	the x-intercept	*X o'qidagi kesilgan nuqta*
52	the ratio of the	*...ning nisbati*
53	the product of	*...ko'paytmasi*
54	is parallel to	*...ga parallel*
55	has no solution	*yechimga ega emas*
56	to the nearest	*...gacha yaxlitlab*
57	at most	*ko'pi bilan*
58	the difference between	*...orasidagi farq*
59	there is	*mavjud*
60	multiplied by	*...ga ko'paytirilgan*

PHRASES – TEZ-TEZ UCHRAYDIGAN IBORALAR

#	English	O'zbekcha
61	corresponding angles	*mos burchaklar*
62	such that	*shundayki*
63	the range of	*...ning qiymatlar sohasi*
64	has infinitely many solutions	*cheksiz ko'p yechimga ega*
65	less than or equal to	*...dan kichik yoki teng*
66	is perpendicular to	*...ga perpendikulyar*
67	per day	*kuniga*
68	must be true	*albatta to'g'ri bo'lishi kerak*
69	no more than	*ko'pi bilan / ...dan oshmaydi*
70	data shown	*keltirilgan ma'lumotlar*
71	vertical angles	*vertikal burchaklar*
72	if and only if	*faqat va faqat ...bo'lgandagina*
73	which of the following could be	*quyidagilardan qaysi biri ...bo'lishi mumkin*
74	is defined as	*...deb ataladi*
75	shown in the table	*jadvalda ko'rsatilgan*
76	at a constant rate	*doimiy sur'atda*
77	the quadratic formula	*kvadrat tenglama ildizlari formulasi*
78	has exactly one solution	*aynan bitta yechimga ega*
79	per year	*yiliga*
80	not equal to	*...ga teng emas*
81	greater than or equal to	*...dan katta yoki teng*
82	decreased by	*...ga kamaytirilgan*
83	a system of equations	*tenglamalar sistemasi*
84	in order to	*...uchun*
85	to the nearest whole number	*eng yaqin butun songacha*
86	is a factor of	*...ning ko'paytuvchisi*
87	solve for	*...ni toping*
88	on average	*o'rtacha hisobda*
89	for all values	*barcha qiymatlar uchun*
90	as shown	*ko'rsatilganidek*

PHRASES – TEZ-TEZ UCHRAYDIGAN IBORALAR

#	English	O'zbekcha
91	alternate interior angles	*almashinuvchi ichki burchaklar*
92	satisfies the equation	*tenglamani qanoatlantiradi*
93	is decreasing	*kamayib bormoqda (kamayuvchi funksiya)*
94	rounded to the nearest	*...gacha yaxlitlangan*
95	the difference of	*...ayirmasi*
96	complete the square	*to'liq kvadratga keltirish*
97	is a linear function	*chiziqli funksiya*
98	as well as	*shuningdek*
99	is one of the	*...lardan biri*
100	at a rate of	*...sur'atda*
101	is increasing	*ortib bormoqda (o'suvchi funksiya)*
102	is a quadratic function	*kvadrat funksiya*
103	could be the value	*qiymati bo'lishi mumkin*
104	shown in the graph	*grafikda ko'rsatilgan*
105	over time	*vaqt o'tishi bilan*
106	is exactly	*aynan ...ga teng*
107	as a function of	*...ning funksiyasi sifatida*
108	reaches a maximum	*maksimumga erishadi*
109	changes by	*...ga o'zgaradi*
110	is a solution of	*...ning yechimi hisoblanadi*
111	with respect to	*...ga nisbatan*
112	the discriminant of	*...ning diskriminanti*
113	in addition to	*...dan tashqari*
114	lies on the line	*chiziq ustida yotadi*
115	is a multiple of	*...ning karralisi*
116	distribute the	*...ni taqsimlang (distributivlik qoidasini qo'llang)*
117	depends on	*...ga bog'liq*
118	combine like terms	*o'xshash hadlarni birlashtiring*
119	which of the following best describes	*quyidagilardan qaysi biri eng yaxshi ifodalaydi*
120	regardless of	*...dan qat'i nazar*

PHRASES – TEZ-TEZ UCHRAYDIGAN IBORALAR

#	English	O'zbekcha
121	is tangent to	*...ga urinadi*
122	is an exponential function	*ko'rsatkichli funksiya*
123	is undefined	*aniqlanmagan*
124	in a given	*berilgan ...da*
125	for some value	*qandaydir qiymat uchun*
126	expressed in terms of	*...orqali ifodalangan*
127	due to	*...sababli*
128	made up of	*...dan tashkil topgan*
129	factor the expression	*ifodani ko'paytuvchilarga ajrating*
130	written in terms of	*...orqali yozilgan*
131	the domain of	*...ning aniqlanish sohasi*
132	substitute the value	*qiymatni o'rniga qo'ying*
133	simplify the expression	*ifodani soddalashtiring*
134	remains constant	*o'zgarmas qoladi*
135	real number solutions	*haqiqiy sonli yechimlar*
136	reaches a minimum	*minimumga erishadi*
137	on the interval	*oraliqda*
138	none of the above	*yuqoridagilarning hech biri*
139	is proportional to	*...ga proporsional*
140	inversely proportional to	*...ga teskari proporsional*
141	in common	*umumiy*
142	exists such that	*shunday ...mavjudki*
143	evaluate the expression	*ifodaning qiymatini hisoblang*
144	directly proportional to	*...ga to'g'ri proporsional*
145	complex solutions	*kompleks yechimlar*
146	as represented	*ifodalanganidek*

Matematikadan qo’rqmang — uni tushunib yeching.

SAT 1200+ seriyasining **1-kitobi** sizni asosiy arifmetikadan boshlab, algebraik ifodalar va koordinatalar tekisligigacha bo’lgan yo’lni bosqichma-bosqich, aniq va tushunarli tarzda o’tkazadi. Har bir dars — formula sahifasi, tushuncha, lug’at va mashqlar bilan to’liq mustahkamlanadi.

- 9 ta tizimli dars, har biri o’z **formula sahifasi** bilan
- **1000+** so’zdan iborat SAT lug’ati — darsma-dars va kitob oxirida to’liq ro’yxat
- Haqiqiy SAT savollar bazasi tahliliga asoslangan mashqlar
- O’zbek tilida tushuntirish, ingliz tilida amaliyot — 2-kitobga tayyor holda o’tasiz

*SAT 1200+ seriyasi: **1-kitob** (Asoslar) → **2-kitob** (Asosiy mavzular) → **3-kitob** (Yuqori ball strategiyalari)*

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